

#1 · Congenital Heart Disease

Name four late complications of the Fontan circulation in adults.

Answer

Protein-losing enteropathy, plastic bronchitis, Fontan-associated liver disease (fibrosis and hepatocellular carcinoma risk), atrial arrhythmias, thromboembolism, and cyanosis from fenestration or collaterals.

#2 · Arrhythmias

What is tachy-brady syndrome, and why does treatment often require a pacemaker?

Answer

Alternating atrial tachyarrhythmias (usually AF) and sinus pauses; rate-controlling drugs worsen the bradycardia, so pacing is placed to permit needed rate or rhythm control therapy.

#3 · Coronary Artery Disease

What two conditions must a patient meet for a plain exercise ECG stress test?

Answer

Ability to exercise to adequate workload and an interpretable baseline ECG without ST-segment abnormalities that obscure ischemia.

#4 · Pulmonary Circulation Disorders

What therapy is reserved for PAH patients with a positive acute vasoreactivity test?

Answer

High-dose calcium channel blockers such as nifedipine, amlodipine, or diltiazem, continued only if a sustained near-normal hemodynamic response is confirmed on follow-up.

#5 · Valvular Disease

What lower INR strategy is permitted for the On-X mechanical aortic valve?

Answer

After 3 months of standard anticoagulation, an INR of 1.5 to 2.0 plus aspirin 75 to 100 mg daily is reasonable.

#6 · Congenital Heart Disease

Which cardiac conditions place pregnancy in modified WHO class IV, where pregnancy is contraindicated?

Answer

Pulmonary arterial hypertension including Eisenmenger syndrome, severe systemic ventricular dysfunction (EF <30%, NYHA III to IV), severe mitral stenosis, severe symptomatic aortic stenosis, severe coarctation, and markedly dilated aortopathy.

#7 · Coronary Artery Disease

What does the GRACE score estimate, and what advantage does it hold over TIMI?

Answer

In-hospital and post-discharge mortality in ACS. It uses continuous variables including hemodynamics, Killip class, creatinine, and cardiac arrest, giving better discrimination than TIMI.

#8 · Systemic Hypertension and Hypotension

What number of drugs and other criteria define resistant hypertension?

Answer

Blood pressure above goal on three optimally dosed agents of different classes including a diuretic, or controlled only on four or more agents, with adherence and white coat effect excluded.

#9 · Arrhythmias

Name three causes of pacemaker failure to capture despite visible pacing spikes.

Answer

Lead dislodgement or fracture, exit block from fibrosis at the electrode-tissue interface, output programmed below threshold, and severe metabolic derangement such as hyperkalemia.

#10 · Normal Cardiovascular Anatomy and Physiology

State the normal right heart catheterization pressures and cardiac index.

Answer

Right atrium 2 to 6 mmHg, mean pulmonary artery below 20 mmHg, wedge pressure 6 to 12 mmHg, and cardiac index 2.5 to 4.0 L/min/m².

#11 · Heart Failure and Cardiomyopathy

What is the target daily dose of bisoprolol in HFrEF?

Answer

10 mg once daily, uptitrated from 1.25 mg daily as tolerated.

#12 · Pulmonary Circulation Disorders

What mean pulmonary artery pressure now defines pulmonary hypertension on right heart catheterization?

Answer

Mean pulmonary artery pressure >20 mmHg at rest, lowered from the historical 25 mmHg threshold in the 2022 ESC/ERS guideline.

#13 · Pericardial Disease

How is constrictive pericarditis distinguished from restrictive cardiomyopathy?

Answer

Constriction preserves or raises medial e' (9 cm/s or higher, with annulus reversus) and near-normal BNP; restriction shows low e' below about 7 cm/s and markedly elevated BNP.

#14 · Heart Failure and Cardiomyopathy

How is transthyretin cardiac amyloidosis diagnosed without a biopsy?

Answer

Grade 2 to 3 myocardial uptake on technetium pyrophosphate scintigraphy plus a negative monoclonal protein screen (serum free light chains and serum and urine immunofixation).

#15 · Pericardial Disease

What is the colchicine regimen and duration for a first episode of acute pericarditis?

Answer

Colchicine 0.5 mg twice daily, reduced to 0.5 mg once daily below 70 kg, continued for 3 months; it roughly halves recurrence.

#16 · Systemic Hypertension and Hypotension

List the classic metabolic side effects of thiazide diuretics.

Answer

Hyponatremia, hypokalemia, hyperuricemia with gout flares, hyperglycemia, and hypercalcemia.

#17 · Heart Failure and Cardiomyopathy

How does an Impella work, and what are key contraindications?

Answer

A catheter-based axial flow pump pulling blood from the LV into the aorta. Contraindicated with LV thrombus, mechanical aortic valve, significant aortic regurgitation, and severe peripheral arterial disease.

#18 · Arrhythmias

On ECG, what finding suggests typical AVNRT rather than orthodromic AVRT during narrow-complex tachycardia?

Answer

A pseudo r' in V1 or pseudo S wave in the inferior leads: the retrograde P is buried in or just after the QRS (very short RP).

#19 · Coronary Artery Disease

What is the default DAPT duration after elective DES placement for chronic coronary disease?

Answer

6 months of aspirin plus clopidogrel, then single antiplatelet therapy.

#20 · Arrhythmias

How do precordial concordance and extreme QRS width help separate VT from SVT with aberrancy?

Answer

All-positive or all-negative QRS across V1 to V6 strongly favors VT, as do QRS >160 ms, a northwest axis, and absence of a typical bundle branch pattern.

#21 · Heart Failure and Cardiomyopathy

What potassium and renal thresholds must be met before starting a mineralocorticoid receptor antagonist?

Answer

Potassium below 5.0 mEq/L and eGFR at least 30 mL/min/1.73 m² at initiation, per the current heart failure guideline.

#22 · Coronary Artery Disease

Describe the de Winter ECG pattern and the occlusion it represents.

Answer

Upsloping ST depression at the J point with tall, symmetric T waves in the precordial leads. It signifies acute proximal LAD occlusion and is treated as a STEMI equivalent.

#23 · Coronary Artery Disease

Clopidogrel for PCI: loading dose, maintenance dose, and its key pharmacogenomic limitation?

Answer

600 mg load, then 75 mg daily. It is a prodrug activated by CYP2C19, so loss-of-function alleles reduce its antiplatelet effect.

#24 · Valvular Disease

For degenerative mitral regurgitation, is repair or replacement preferred?

Answer

Repair, when a durable result is likely; asymptomatic patients should be referred to experienced valve centers where successful repair likelihood exceeds 95%.

#25 · Heart Failure and Cardiomyopathy

What ejection fraction cutoff defines heart failure with preserved ejection fraction (HFpEF)?

Answer

LVEF $\geq 50\%$ defines HFpEF, together with symptoms and evidence of elevated filling pressures such as raised natriuretic peptides or diastolic dysfunction.

#26 · Coronary Artery Disease

State the Duke treadmill score formula and the score defining high risk.

Answer

Exercise minutes minus $5 \times$ maximal ST deviation in mm minus $4 \times$ angina index. A score ≤ -11 is high risk; $\geq +5$ is low risk.

#27 · Arrhythmias

Which oral antiarrhythmic legally requires 3 days of inpatient initiation with ECG monitoring, and why?

Answer

Dofetilide, because of dose-related torsades risk; dosing is set by creatinine clearance and the QTc response after each of the first doses. It is contraindicated if QTc >440 ms at baseline.

#28 · Arrhythmias

Preexcited atrial fibrillation: which drug classes are contraindicated, and what should be given instead?

Answer

Avoid AV nodal blockers (adenosine, verapamil, diltiazem, beta blockers, digoxin, IV amiodarone); they can accelerate conduction to VF. Use IV procainamide or ibutilide, or cardiovert if unstable.

#29 · Valvular Disease

What is the Austin Flint murmur and how is it distinguished from mitral stenosis?

Answer

A low-pitched apical mid-diastolic rumble in severe aortic regurgitation, from the jet striking the anterior mitral leaflet; unlike mitral stenosis there is no opening snap or loud S1.

#30 · Coronary Artery Disease

Define the four Killip classes in acute MI.

Answer

Class I, no heart failure; II, rales or S3; III, frank pulmonary edema; IV, cardiogenic shock. Mortality rises steeply with each class.

#31 · Coronary Artery Disease

Ventricular septal rupture after MI: timing, exam finding, and the confirmatory hemodynamic clue?

Answer

3 to 5 days post-MI, a new harsh holosystolic murmur often with a palpable thrill. Right heart catheterization shows an oxygen saturation step-up at the right ventricle.

#32 · Coronary Artery Disease

Which two characteristic side effects of ticagrelor are not caused by the other oral P2Y₁₂ inhibitors?

Answer

Dyspnea without pulmonary pathology and ventricular pauses, both adenosine mediated. Dyspnea causes discontinuation in some patients.

#33 · Coronary Artery Disease

What anterior-lead ECG clues suggest posterior MI, and how is it confirmed?

Answer

ST depression in V1 to V3 with tall R waves and upright T waves. Confirm with posterior leads V7 to V9 showing ST elevation ≥ 0.5 mm.

#34 · Valvular Disease

What INR target applies to a mechanical mitral valve, or a mechanical aortic valve with risk factors?

Answer

An INR of 3.0, range 2.5 to 3.5, for mitral prostheses, older-generation valves, or aortic prostheses with thromboembolic risk factors such as atrial fibrillation or prior embolism.

#35 · Coronary Artery Disease

ST elevation in leads I, aVL, V5, and V6 points to which territory and vessels?

Answer

Lateral wall, supplied by the left circumflex or a diagonal branch of the LAD.

#36 · Normal Cardiovascular Anatomy and Physiology

Define contractility and name two things that increase it.

Answer

The intrinsic force of myocardial contraction independent of preload and afterload. Catecholamines and digoxin increase it; beta blockers and hypoxia reduce it.

#37 · Systemic Hypertension and Hypotension

What is droxidopa and when is it used?

Answer

An oral norepinephrine prodrug approved for symptomatic neurogenic orthostatic hypotension, as in Parkinson disease or pure autonomic failure. It shares the supine hypertension caution.

#38 · Congenital Heart Disease

What gradient threshold supports intervening on native or recurrent aortic coarctation?

Answer

A peak-to-peak coarctation gradient ≥ 20 mmHg with hypertension, or a lesser gradient with significant collaterals or anatomic narrowing plus hypertension. Stenting is favored in suitable adult anatomy.

#39 · Pericardial Disease

Describe the stage 1 ECG of acute pericarditis.

Answer

Diffuse concave ST elevation with PR-segment depression across most leads, plus reciprocal ST depression and PR elevation in aVR.

#40 · Congenital Heart Disease

What findings justify closing a ventricular septal defect in an adult?

Answer

A hemodynamically significant shunt ($Q_p:Q_s \geq 1.5:1$) with left ventricular volume overload, or a history of endocarditis, provided pulmonary vascular disease is not severe.

#41 · Valvular Disease

What is the most common cause of infective endocarditis overall, and its classic settings?

Answer

Staphylococcus aureus, causing acute, rapidly destructive disease; it dominates injection drug use endocarditis, which favors the tricuspid valve.

#42 · Systemic Disorders Affecting the Circulatory System

Which cardiovascular conditions make an SGLT2 inhibitor guideline-directed therapy?

Answer

Heart failure across the ejection fraction spectrum, type 2 diabetes with established atherosclerotic disease or high risk, and CKD, reducing heart failure hospitalization and cardiovascular death.

#43 · Vascular Diseases

What ankle-brachial index value diagnoses peripheral artery disease?

Answer

≤ 0.90 at rest. Values 0.91 to 0.99 are borderline; exercise ABI can unmask disease when resting values are borderline with classic symptoms.

#44 · Coronary Artery Disease

Which NSTEMI-ACS features mandate an immediate invasive strategy, within 2 hours?

Answer

Refractory angina, hemodynamic instability or cardiogenic shock, electrical instability with sustained VT or VF, and acute heart failure attributable to ischemia.

#45 · Valvular Disease

What are the classic physical findings of severe tricuspid regurgitation?

Answer

Prominent c-v waves in the jugular venous pulse, a pulsatile liver, and a holosystolic lower left sternal border murmur that augments with inspiration.

#46 · Systemic Disorders Affecting the Circulatory System

What is Libman-Sacks endocarditis, and which valve does it most often involve?

Answer

Sterile verrucous vegetations on valve surfaces, most often mitral, seen in SLE and antiphospholipid syndrome. It can cause regurgitation and embolic stroke despite negative blood cultures.

#47 · Vascular Diseases

List the six Ps of acute limb ischemia.

Answer

Pain, pallor, pulselessness, paresthesia, paralysis, and poikilothermia (coolness).

#48 · Vascular Diseases

How is a lower-extremity superficial vein thrombosis of ≥ 5 cm treated?

Answer

Fondaparinux 2.5 mg daily for 45 days. Clots within 3 cm of the saphenofemoral junction are managed like deep vein thrombosis with full anticoagulation.

#49 · Arrhythmias

New atrial fibrillation after cardiac surgery: when does it peak, and what is the anticoagulation implication?

Answer

Peaks on postoperative days 2 to 4 and is often self-limited, but it carries later stroke and AF recurrence risk, so anticoagulation is reasonable when CHA₂DS₂-VASc is elevated.

#50 · Arrhythmias

Which idiopathic VT presents with RBBB morphology plus left axis and responds to verapamil?

Answer

Left posterior fascicular (Belhassen) VT, a reentrant rhythm using the Purkinje network in a structurally normal heart. IV verapamil terminates it.

#51 · Valvular Disease

What is the strongest indication for early surgery in infective endocarditis?

Answer

Valve dysfunction causing heart failure; early surgery during the initial hospitalization is class 1.

#52 · Coronary Artery Disease

How does the Canadian Cardiovascular Society define class III angina?

Answer

Marked limitation of ordinary activity: angina on walking one to two level blocks or climbing one flight of stairs at a normal pace.

#53 · Heart Failure and Cardiomyopathy

What ejection fraction range defines heart failure with mildly reduced ejection fraction (HFmrEF)?

Answer

LVEF 41 to 49% defines HFmrEF, with evidence of elevated filling pressures supporting the diagnosis.

#54 · Coronary Artery Disease

Enoxaparin dosing in NSTEMI-ACS, and the adjustment for creatinine clearance below 30 mL/min?

Answer

1 mg/kg subcutaneously every 12 hours. Reduce to 1 mg/kg once daily when creatinine clearance is below 30 mL/min.

#55 · Heart Failure and Cardiomyopathy

What defines takotsubo (stress) cardiomyopathy, and what is its usual course?

Answer

Transient regional wall motion abnormality, classically apical ballooning, extending beyond a single coronary territory, often after emotional or physical stress, without culprit coronary occlusion. LVEF typically recovers within days to weeks.

#56 · Heart Failure and Cardiomyopathy

Why are verapamil and diltiazem contraindicated in HFrEF?

Answer

Non-dihydropyridine calcium channel blockers are negative inotropes that worsen outcomes in reduced ejection fraction; amlodipine is the safer option when a CCB is needed.

#57 · Arrhythmias

What pretreatment reduces torsades risk when using ibutilide to convert atrial flutter or fibrillation?

Answer

IV magnesium before infusion, plus normal potassium and 4 hours of continuous ECG monitoring afterward; torsades occurs in several percent of recipients.

#58 · Valvular Disease

Which cardiac conditions qualify for antibiotic endocarditis prophylaxis?

Answer

Prosthetic valves or prosthetic valve material including TAVR, prior endocarditis, unrepaired cyanotic congenital disease, repairs with prosthetic material within 6 months or with residual defects, and transplant valvulopathy.

#59 · Vascular Diseases

What clinical features raise suspicion for giant cell arteritis?

Answer

Age ≥ 50 with new headache, jaw claudication, scalp tenderness, visual symptoms, polymyalgia rheumatica, and elevated ESR or CRP.

#60 · Vascular Diseases

What is the classic presentation of acute embolic mesenteric ischemia?

Answer

Sudden severe abdominal pain out of proportion to examination, often with atrial fibrillation, from an embolus lodging in the superior mesenteric artery. Lactate elevation is a late sign.

#61 · Systemic Disorders Affecting the Circulatory System

How does trastuzumab cardiotoxicity differ from anthracycline toxicity, and how is it monitored?

Answer

It is not dose-dependent and is usually reversible after interruption. Monitor LVEF at baseline and every 3 months during therapy.

#62 · Coronary Artery Disease

Free wall rupture after MI: timing window and classic presentation?

Answer

Most occur within the first 5 to 14 days. It presents as sudden tamponade with pulseless electrical activity and is usually fatal without emergency surgery.

#63 · Systemic Hypertension and Hypotension

Which antihypertensive class is preferred in CKD with albuminuria?

Answer

An ACE inhibitor or ARB, which slows progression of proteinuric kidney disease. Never combine the two classes.

#64 · Arrhythmias

What defines complete heart block on the ECG strip?

Answer

AV dissociation with an atrial rate faster than a regular ventricular escape rhythm; no P wave conducts. Escape is junctional (narrow, 40 to 60) or ventricular (wide, 20 to 40).

#65 · Valvular Disease

What defines very severe aortic stenosis, and why does it matter in asymptomatic patients?

Answer

Peak velocity ≥ 5 m/s or mean gradient ≥ 60 mmHg; in asymptomatic patients at low surgical risk, intervention is reasonable at this severity.

#66 · Valvular Disease

Why does acute severe aortic regurgitation look different from chronic, and how is it managed?

Answer

The unadapted ventricle produces a soft S1, short murmur, and pulmonary edema without wide pulse pressure; treatment is emergency surgery, and an intra-aortic balloon pump is contraindicated.

#67 · Valvular Disease

What triggers surgery in asymptomatic chronic severe aortic regurgitation?

Answer

LVEF $\leq 55\%$ is a class 1 trigger; surgery is also reasonable when LVESD exceeds 50 mm (indexed above 25 mm/m²) or during other cardiac surgery.

#68 · Coronary Artery Disease

Papillary muscle rupture after MI: typical timing, which muscle, and why?

Answer

3 to 5 days post-MI, usually the posteromedial papillary muscle because it has a single blood supply from the posterior descending artery. It causes acute severe MR with pulmonary edema.

#69 · Vascular Diseases

How does a type A aortic dissection differ from type B in management?

Answer

Type A involves the ascending aorta and requires emergency surgery. Type B spares the ascending aorta and is managed medically unless complicated, when TEVAR is preferred.

#70 · Coronary Artery Disease

What LDL-C reduction defines moderate-intensity statin therapy, with two example regimens?

Answer

30 to 49% reduction, such as atorvastatin 10 to 20 mg or rosuvastatin 5 to 10 mg daily. Simvastatin 20 to 40 mg also qualifies.

#71 · Heart Failure and Cardiomyopathy

How is the cold and wet acute heart failure profile managed?

Answer

Congested and hypoperfused: cautious diuresis plus inotropic support, with vasopressors for hypotension and early consideration of invasive hemodynamics and mechanical circulatory support.

#72 · Pericardial Disease

In what order do echocardiographic signs of tamponade appear as pressure rises?

Answer

A dilated inferior vena cava without inspiratory collapse appears first (sensitive but not specific), then right atrial collapse lasting more than a third of the cardiac cycle, then early diastolic right ventricular free wall collapse (most specific).

#73 · Heart Failure and Cardiomyopathy

What are the age-stratified NT-proBNP cutoffs for ruling in acute heart failure?

Answer

>450 pg/mL under age 50, >900 pg/mL for ages 50 to 75, and >1800 pg/mL over age 75.

#74 · Arrhythmias

Describe the diagnostic type 1 Brugada ECG pattern and where the leads may need to move.

Answer

Coved ST elevation ≥ 2 mm with T inversion in V1 or V2; sensitivity improves with leads placed in the 2nd or 3rd intercostal space.

#75 · Coronary Artery Disease

What is the default DAPT duration after ACS, regardless of stenting?

Answer

12 months of aspirin plus a P2Y₁₂ inhibitor, with ticagrelor or prasugrel generally preferred over clopidogrel after PCI.

#76 · Coronary Artery Disease

ST elevation in leads V1 through V4 localizes to which territory and artery?

Answer

Anterior wall and septum, supplied by the left anterior descending artery.

#77 · Coronary Artery Disease

How is SCAD usually managed, and why is fibrinolysis dangerous?

Answer

Conservatively when flow is preserved and the patient is stable, since most dissections heal spontaneously and PCI carries high complication rates. Fibrinolysis can extend the intramural hematoma.

#78 · Valvular Disease

Who is a candidate for percutaneous balloon mitral commissurotomy?

Answer

Symptomatic patients with severe rheumatic mitral stenosis and favorable valve morphology, classically a pliable, noncalcified valve with a Wilkins score of 8 or less.

#79 · Coronary Artery Disease

How many days before elective CABG should clopidogrel be held?

Answer

At least 5 days, to allow platelet function recovery from irreversible P2Y₁₂ inhibition.

#80 · Coronary Artery Disease

Which proton pump inhibitors most reduce clopidogrel activation, and what is the safer alternative?

Answer

Omeprazole and esomeprazole, through CYP2C19 inhibition. Pantoprazole has minimal interaction and is preferred when a PPI is needed.

#81 · Arrhythmias

Why must flecainide and propafenone be avoided in coronary artery disease or structural heart disease?

Answer

The CAST trial showed class IC agents increase mortality after myocardial infarction by proarrhythmia. They are reserved for structurally normal hearts.

#82 · Arrhythmias

In a patient with prior myocardial infarction, what is the default diagnosis for wide-complex tachycardia?

Answer

Ventricular tachycardia until proven otherwise; structural heart disease makes VT far more likely than SVT with aberrancy. When unsure, treat as VT.

#83 · Heart Failure and Cardiomyopathy

What is the target dose of carvedilol in HFrEF for most adults?

Answer

25 mg twice daily; 50 mg twice daily may be used in patients weighing more than 85 kg.

#84 · Valvular Disease

On dobutamine stress echo for low-flow, low-gradient aortic stenosis, what distinguishes true severe from pseudosevere disease?

Answer

True severe: velocity rises to ≥ 4 m/s while valve area stays ≤ 1.0 cm². Pseudosevere: valve area opens above 1.0 cm² as flow increases.

#85 · Heart Failure and Cardiomyopathy

When should potassium and renal function be rechecked after starting or uptitrating an MRA?

Answer

Within 1 to 2 weeks of initiation or dose change, then periodically thereafter. Stop the MRA if potassium exceeds 5.5 mEq/L or renal function deteriorates significantly.

#86 · Pericardial Disease

What is the significance of tuberculous pericarditis worldwide, and its treatment?

Answer

It is the leading cause of constrictive pericarditis in endemic regions; treatment is standard four-drug antituberculous therapy, with pericardiectomy for constriction despite treatment.

#87 · Heart Failure and Cardiomyopathy

List the major sudden cardiac death risk markers that support ICD consideration in HCM.

Answer

Family history of HCM sudden death, massive hypertrophy ≥ 30 mm, unexplained syncope, nonsustained VT, apical aneurysm, extensive late gadolinium enhancement, and LVEF $< 50\%$.

#88 · Coronary Artery Disease

What distinguishes a Type 1 from a Type 2 myocardial infarction?

Answer

Type 1 is atherothrombotic, caused by plaque rupture or erosion with thrombus. Type 2 is oxygen supply-demand mismatch without acute atherothrombosis, such as severe anemia, tachyarrhythmia, or hypotension.

#89 · Coronary Artery Disease

In NSTEMI-ACS, a GRACE score above what value supports an early invasive strategy within 24 hours?

Answer

GRACE score above 140. High-risk patients benefit from angiography within 24 hours of presentation.

#90 · Valvular Disease

When does suspected endocarditis call for transesophageal rather than only transthoracic echocardiography?

Answer

When TTE is nondiagnostic, with a prosthetic valve or intracardiac device, with *Staphylococcus aureus* bacteremia, or when intracardiac complications such as abscess are suspected.

#91 · Valvular Disease

How is suspected bioprosthetic valve thrombosis treated in a hemodynamically stable patient?

Answer

With a trial of vitamin K antagonist anticoagulation; gradients and leaflet thickening often improve without surgery.

#92 · Arrhythmias

What does adenosine do to atrial flutter, and why is that diagnostically useful?

Answer

It transiently blocks the AV node, slowing the ventricular response and unmasking flutter waves; it does not terminate the flutter circuit itself.

#93 · Heart Failure and Cardiomyopathy

When should giant cell myocarditis be suspected, and how is it managed?

Answer

Fulminant heart failure with ventricular arrhythmias or heart block in a young or middle-aged adult. Confirm with endomyocardial biopsy; treat with combination immunosuppression and early transplant or MCS evaluation.

#94 · Vascular Diseases

At what aortic diameter is repair advised in Marfan syndrome?

Answer

≥5.0 cm at the root or ascending aorta, and lower (around 4.5 cm) with family history of dissection, rapid growth, or planned pregnancy.

#95 · Arrhythmias

In the 2023 ACC/AHA/ACCP/HRS atrial fibrillation guideline, what does Stage 3 represent in the AF staging scheme?

Answer

Clinically manifest AF (paroxysmal, persistent, long-standing persistent, or successfully ablated). Stage 1 is at risk, Stage 2 is pre-AF, Stage 4 is permanent AF.

#96 · Heart Failure and Cardiomyopathy

What hemodynamic criteria define cardiogenic shock on right heart catheterization?

Answer

Cardiac index <2.2 L/min/m² with pulmonary capillary wedge pressure ≥ 15 mmHg, plus systemic hypotension or hypoperfusion.

#97 · Valvular Disease

What defines classical low-flow, low-gradient severe aortic stenosis?

Answer

Valve area ≤ 1.0 cm² with peak velocity below 4 m/s and mean gradient below 40 mmHg in a patient with LVEF below 50%.

#98 · Arrhythmias

Which beta blockers are preferred in congenital long QT syndrome, and which everyday exposures must patients avoid?

Answer

Nadolol or propranolol (nonselective agents have the best data). Avoid all QT-prolonging drugs (crediblemeds list) and correct hypokalemia and hypomagnesemia.

#99 · Heart Failure and Cardiomyopathy

What is the standard strategy for loop diuretic resistance in congested heart failure?

Answer

Add a thiazide-type diuretic such as metolazone or chlorothiazide for sequential nephron blockade, blocking distal sodium reabsorption upregulated by chronic loop use. Monitor potassium closely.

#100 · Coronary Artery Disease

STEMI at a PCI-capable hospital: what is the goal first medical contact to device time?

Answer

90 minutes or less from first medical contact to device.

#101 · Congenital Heart Disease

In repaired tetralogy with severe pulmonary regurgitation, what triggers pulmonary valve replacement in an asymptomatic patient?

Answer

Progressive RV dilation (commonly RVEDVi at least 160 mL/m² or RVESVi at least 80 mL/m²), RV systolic dysfunction, or important tricuspid regurgitation.

#102 · Arrhythmias

Who qualifies for a secondary prevention ICD without any waiting period or EF threshold?

Answer

Survivors of VF or hemodynamically unstable sustained VT, and those with sustained VT plus structural heart disease, once fully reversible causes are excluded.

#103 · Heart Failure and Cardiomyopathy

How is cardiac power output calculated, and what value predicts mortality in cardiogenic shock?

Answer

CPO = mean arterial pressure × cardiac output / 451. Values <0.6 watts strongly predict in-hospital mortality.

#104 · Systemic Hypertension and Hypotension

Which oral antihypertensive agents are preferred for chronic hypertension in pregnancy?

Answer

Labetalol, extended-release nifedipine, and methyldopa. For severe hypertension in pregnancy, IV labetalol, IV hydralazine, or oral nifedipine are used.

#105 · Valvular Disease

How long does rheumatic fever prophylaxis continue by carditis status?

Answer

Carditis with residual valve disease: 10 years or until age 40, whichever is longer. Carditis without residual disease: 10 years or until 21. No carditis: 5 years or until 21.

#106 · Vascular Diseases

Why must beta blockade precede vasodilators in aortic dissection anti-impulse therapy?

Answer

A vasodilator alone triggers reflex tachycardia and raises aortic wall shear stress (dP/dt), extending the dissection. Control rate first, then add the vasodilator.

#107 · Coronary Artery Disease

How is Type 4a, PCI-related myocardial infarction, defined by troponin?

Answer

Troponin rise to more than 5× the 99th percentile within 48 hours of PCI in patients with normal baseline, plus evidence of new ischemia.

#108 · Arrhythmias

What arrhythmia trigger and gene characterize long QT type 1, and which activity is classically dangerous?

Answer

Exercise, especially swimming and diving, triggers events; the gene is KCNQ1 (slow potassium channel loss of function). Beta blockers are most protective in LQT1.

#109 · Congenital Heart Disease

Define the anatomic hallmark of Ebstein anomaly and its two classic associations.

Answer

Apical displacement of the septal tricuspid leaflet with atrialization of the right ventricle and tricuspid regurgitation; associated with accessory pathways (WPW) and an interatrial shunt (ASD or PFO).

#110 · Coronary Artery Disease

When is dobutamine the stress agent of choice, and when should it be avoided?

Answer

When the patient cannot exercise and vasodilators are contraindicated, such as severe bronchospasm. Avoid it with serious ventricular arrhythmias, recent ACS, or severe LVOT obstruction.

#111 · Systemic Hypertension and Hypotension

What blood pressure range defines the elevated category?

Answer

Systolic 120 to 129 mmHg with diastolic below 80 mmHg.

#112 · Systemic Hypertension and Hypotension

What are the blood pressure and heart rate targets in acute aortic dissection, and the drug sequence?

Answer

Heart rate below 60 and systolic 100 to 120 mmHg within about 20 minutes. Give an IV beta blocker such as esmolol first, then add a vasodilator if needed.

#113 · Coronary Artery Disease

Icosapent ethyl: dose, the triglyceride range it is indicated for, and its key risks?

Answer

2 g twice daily for triglycerides 135 to 499 mg/dL on a statin with ASCVD or diabetes plus risk factors, per REDUCE-IT. Risks include atrial fibrillation and bleeding.

#114 · Coronary Artery Disease

Prasugrel dosing and the patients in whom it is contraindicated or requires caution?

Answer

60 mg load, then 10 mg daily. Contraindicated with prior stroke or TIA; avoid or use 5 mg if age ≥ 75 years or weight under 60 kg.

#115 · Systemic Hypertension and Hypotension

Which antihypertensives are indicated after myocardial infarction or in HFrEF?

Answer

A beta blocker plus an ACE inhibitor or ARB, for survival benefit beyond blood pressure lowering.

#116 · Arrhythmias

Why is a waiting period on GDMT required before a primary prevention ICD in new nonischemic cardiomyopathy?

Answer

At least 3 months of guideline-directed therapy is required because EF often improves above 35%, removing the indication. DANISH tempered mortality expectations in nonischemic disease.

#117 · Coronary Artery Disease

When should oral beta blockers start after MI, and what findings should delay them?

Answer

Within the first 24 hours unless there are signs of heart failure, low-output state, risk of cardiogenic shock, high-grade AV block, or active bronchospasm.

#118 · Pericardial Disease

What is the role of corticosteroids in acute pericarditis?

Answer

Not first line because they raise recurrence risk; use low dose, 0.2 to 0.5 mg/kg prednisone with a slow taper, for NSAID contraindication or failure, autoimmune or uremic disease, or pregnancy.

#119 · Heart Failure and Cardiomyopathy

How can potassium binders help maintain guideline-directed therapy in HFrEF with recurrent hyperkalemia?

Answer

Patiromer or sodium zirconium cyclosilicate can lower potassium enough to allow continued RAAS inhibitor and MRA therapy rather than permanent discontinuation.

#120 · Valvular Disease

Describe the classic aortic stenosis murmur and the Gallavardin phenomenon.

Answer

Harsh crescendo-decrescendo systolic murmur at the right upper sternal border radiating to the carotids; the Gallavardin phenomenon is a musical apical radiation mimicking mitral regurgitation.

#121 • Valvular Disease

For patients over 65 needing aortic valve replacement, which prosthesis type does the guideline favor?

Answer

A bioprosthesis is reasonable past age 70, since structural deterioration is slower at older ages and lifelong anticoagulation is avoided. Age 50-70 is shared decision-making, and under 50 favors mechanical.

#122 • Coronary Artery Disease

Sgarbossa criteria for MI in LBBB: list the three findings and their points.

Answer

Concordant ST elevation ≥ 1 mm scores 5, concordant ST depression ≥ 1 mm in V1 to V3 scores 3, discordant ST elevation ≥ 5 mm scores 2. A total ≥ 3 is diagnostic.

#123 · Coronary Artery Disease

A PRECISE-DAPT score at what threshold identifies high bleeding risk and favors shorter DAPT?

Answer

≥25. These patients gain no ischemic benefit from prolonged DAPT but bleed substantially more.

#124 · Arrhythmias

In which airway disease is adenosine best avoided, and what other setting blunts or exaggerates its effect?

Answer

Avoid in severe reactive airway disease (bronchospasm risk). Dipyridamole potentiates adenosine; methylxanthines like theophylline and caffeine antagonize it. Heart transplant recipients are hypersensitive.

#125 · Valvular Disease

When is surgery indicated for symptomatic severe aortic regurgitation?

Answer

Symptoms alone are a class 1 indication for aortic valve surgery regardless of left ventricular systolic function.

#126 · Coronary Artery Disease

Name the high-intensity statin regimens and the expected LDL-C reduction.

Answer

Atorvastatin 40 to 80 mg or rosuvastatin 20 to 40 mg daily, lowering LDL-C by $\geq 50\%$.

#127 · Congenital Heart Disease

Which valve abnormality accompanies aortic coarctation in over half of patients, and what vascular lesion also associates?

Answer

Bicuspid aortic valve, in roughly 60%. Intracranial (berry) aneurysms also occur at increased frequency, and ascending aortopathy warrants surveillance.

#128 · Congenital Heart Disease

What venous anomaly travels with a sinus venosus ASD, and why can it not be device closed?

Answer

Partial anomalous pulmonary venous return, usually the right upper pulmonary vein with a superior sinus venosus defect. The defect lacks rims and involves the veins, so repair is surgical.

#129 · Pericardial Disease

How is purulent bacterial pericarditis managed, and why is drainage mandatory?

Answer

Urgent pericardial drainage plus intravenous antibiotics; drainage is mandatory because antibiotics alone fail and mortality untreated approaches 100%.

#130 · Vascular Diseases

What is the treatment rule when giant cell arteritis is suspected?

Answer

Start high-dose glucocorticoids immediately; do not wait for biopsy. Temporal artery biopsy remains diagnostic for up to about 2 weeks after starting steroids.

#131 · Coronary Artery Disease

Ezetimibe: mechanism and typical additional LDL-C lowering on top of a statin?

Answer

Blocks the NPC1L1 transporter, inhibiting intestinal cholesterol absorption, and lowers LDL-C by roughly 18 to 25%.

#132 · Arrhythmias

What is the most feared toxicity of long-term amiodarone, and how does it present?

Answer

Pulmonary toxicity: subacute cough, dyspnea, and infiltrates with reduced DLCO. Stop the drug; corticosteroids are used in significant cases. It can be fatal if missed.

#133 · Pericardial Disease

What is transient constrictive pericarditis, and how does it change management?

Answer

Constriction with active inflammation that can resolve; a 2 to 3 month trial of anti-inflammatory therapy is warranted in stable patients before committing to pericardiectomy.

#134 · Heart Failure and Cardiomyopathy

What hemodynamic problem does VA-ECMO create for the left ventricle, and how is it addressed?

Answer

Retrograde aortic flow raises LV afterload, causing LV distension and pulmonary edema. Manage with LV venting or unloading, such as an Impella (ECPELLA) or atrial septostomy.

#135 · Coronary Artery Disease

Define MINOCA and name the single most useful test to establish its mechanism.

Answer

Myocardial infarction meeting the universal definition with no coronary stenosis $\geq 50\%$ and no alternative diagnosis. Cardiac MRI is the key test, separating true infarction from myocarditis.

#136 · Coronary Artery Disease

How many days before elective CABG should ticagrelor be held?

Answer

At least 3 days. Although reversible, its twice-daily dosing and active metabolite still impair platelets for several days.

#137 · Heart Failure and Cardiomyopathy

What is disopyramide's role in obstructive hypertrophic cardiomyopathy?

Answer

A negative inotrope added to a beta-blocker or calcium channel blocker for persistent obstructive symptoms. Its vagolytic side effects and QT prolongation require monitoring.

#138 · Heart Failure and Cardiomyopathy

Which antiarrhythmic drugs are contraindicated or harmful in HFrEF?

Answer

Class IC agents such as flecainide and propafenone increase mortality in structural heart disease, and dronedarone is contraindicated in NYHA class III to IV or recently decompensated heart failure.

#139 · Coronary Artery Disease

Nitroglycerin is contraindicated within what interval after each phosphodiesterase-5 inhibitor?

Answer

24 hours after sildenafil or vardenafil and 48 hours after tadalafil, because combined use can cause refractory hypotension.

#140 · Valvular Disease

What separates primary from secondary mitral regurgitation, and why does the distinction drive therapy?

Answer

Primary MR is intrinsic leaflet or chordal disease treated with valve intervention; secondary MR reflects ventricular or atrial disease with structurally normal leaflets, treated first with guideline-directed medical therapy.

#141 · Heart Failure and Cardiomyopathy

How long must an ACE inhibitor be held before starting sacubitril/valsartan?

Answer

36 hours. The washout separates ACE inhibition from neprilysin inhibition before the first ARNI dose.

#142 · Vascular Diseases

Give the surveillance intervals for an abdominal aortic aneurysm below repair threshold.

Answer

Roughly every 3 years for 3.0 to 3.9 cm, every 12 months for 4.0 to 4.9 cm, and every 6 months for 5.0 to 5.4 cm.

#143 · Valvular Disease

What defines contractile (flow) reserve on dobutamine testing, and what does its absence mean?

Answer

A stroke volume increase of $\geq 20\%$. Absence predicts higher operative risk, though valve replacement can still improve outcomes in true severe stenosis.

#144 · Valvular Disease

In the 2020 ACC/AHA framework, which aortic stenosis patients should preferentially receive surgical AVR by age?

Answer

Patients under 65, or with life expectancy over 20 years, are preferentially treated with surgical aortic valve replacement.

#145 · Systemic Hypertension and Hypotension

What exam findings suggest aortic coarctation in a hypertensive patient?

Answer

Higher arm than leg blood pressure, radiofemoral pulse delay, and a systolic murmur over the back. Rib notching may appear on chest radiograph.

#146 · Heart Failure and Cardiomyopathy

Summarize the five SCAI cardiogenic shock stages A through E.

Answer

A: at risk. B: beginning, hypotension or tachycardia without hypoperfusion. C: classic, hypoperfusion requiring intervention. D: deteriorating despite initial support. E: extremis, circulatory collapse or refractory arrest.

#147 • Coronary Artery Disease

Which baseline ECG findings make exercise ST segments uninterpretable and require stress imaging?

Answer

LBBB, ventricular pacing, pre-excitation, LVH with repolarization abnormality, digoxin effect, and resting ST depression ≥ 1 mm.

#148 • Congenital Heart Disease

Give three management principles in Eisenmenger syndrome, including the modern view of phlebotomy.

Answer

Avoid pregnancy, dehydration, and systemic vasodilators; use air filters on IV lines; consider pulmonary vasodilators (bosentan, per BREATHE-5). Phlebotomize only for symptomatic hyperviscosity, never for hematocrit alone.

#149 · Congenital Heart Disease

What does differential cyanosis (blue toes, pink fingers) indicate in an adult with a PDA?

Answer

Eisenmenger physiology of the ductus: desaturated blood shunts right-to-left into the descending aorta beyond the left subclavian, sparing the upper body. Closure is then contraindicated.

#150 · Heart Failure and Cardiomyopathy

How does trastuzumab cardiotoxicity differ from anthracycline cardiotoxicity, and how is it monitored?

Answer

Largely reversible and not cumulative-dose dependent. Monitor LVEF about every 3 months during therapy, holding trastuzumab for significant declines and rechallenging after recovery.

#151 · Vascular Diseases

What features distinguish primary from secondary Raynaud phenomenon?

Answer

Primary is symmetric, starts young, and spares tissue with normal nailfold capillaries. Secondary suggests connective tissue disease: onset after 40, asymmetry, digital ulcers, abnormal nailfold capillaroscopy, positive ANA.

#152 · Systemic Disorders Affecting the Circulatory System

How does immune checkpoint inhibitor myocarditis present and how is it treated?

Answer

Early, at a median of about 30 days after starting therapy, often with arrhythmia or heart block and high fatality. Stop the drug and give high-dose methylprednisolone.

#153 · Coronary Artery Disease

Sublingual nitroglycerin for angina: dose, repeat schedule, and when to call emergency services?

Answer

0.3 to 0.4 mg, repeated every 5 minutes for up to 3 doses. Call 911 if pain is unimproved 5 minutes after the first dose.

#154 · Valvular Disease

What is the standard regimen for rheumatic fever secondary prophylaxis?

Answer

Benzathine penicillin G 1.2 million units intramuscularly every 4 weeks, shortened to every 3 weeks in high-risk settings.

#155 · Vascular Diseases

What antithrombotic combination reduces events in chronic symptomatic PAD?

Answer

Rivaroxaban 2.5 mg twice daily plus aspirin 81 mg, which lowered cardiovascular and major adverse limb events in COMPASS and after revascularization in VOYAGER PAD.

#156 · Heart Failure and Cardiomyopathy

What left ventricular ejection fraction defines heart failure with reduced ejection fraction (HFrEF)?

Answer

LVEF $\leq 40\%$ defines HFrEF in the 2022 AHA/ACC/HFSA guideline classification.

#157 · Arrhythmias

Which long QT subtype is triggered by sudden auditory stimuli and the postpartum period?

Answer

LQT2, from KCNH2 (hERG) potassium channel loss of function. Alarm clocks and phone rings are the classic triggers.

#158 · Congenital Heart Disease

What is the dominant residual lesion after tetralogy of Fallot repair, and how is it surveilled?

Answer

Chronic severe pulmonary regurgitation from the transannular patch, causing progressive RV dilation. Surveillance uses periodic cardiac MRI to quantify RV volumes and function, plus ECG and Holter.

#159 · Heart Failure and Cardiomyopathy

Which common condition lowers natriuretic peptide levels and can mask heart failure?

Answer

Obesity. Natriuretic peptide levels are disproportionately low in obese patients, so heart failure can be present despite modest values.

#160 · Congenital Heart Disease

Describe the classic murmur and peripheral pulse findings of a patent ductus arteriosus.

Answer

A continuous machinery murmur, loudest at the left infraclavicular area, with bounding pulses and a wide pulse pressure from aortic diastolic runoff.

#161 · Valvular Disease

What is the INR target for a current-generation mechanical aortic valve without risk factors?

Answer

An INR of 2.5, range 2.0 to 3.0, for bileaflet or current-generation single-tilting-disc aortic prostheses.

#162 · Valvular Disease

What features characterize fungal endocarditis and its management?

Answer

Candida or Aspergillus infection with large, embolization-prone vegetations, usually in patients with lines, immunosuppression, or injection drug use; treatment combines surgery with prolonged antifungal therapy.

#163 · Vascular Diseases

When is revascularization justified for atherosclerotic renal artery stenosis?

Answer

CORAL and ASTRAL showed no routine benefit over medical therapy. Reserve stenting for recurrent flash pulmonary edema, refractory hypertension, or progressive renal failure from bilateral stenosis.

#164 · Pulmonary Circulation Disorders

Outline the treatment ladder for chronic thromboembolic pulmonary hypertension.

Answer

Lifelong anticoagulation for all. Pulmonary endarterectomy is first-line for operable disease; balloon pulmonary angioplasty and riociguat treat inoperable or persistent post-surgical disease.

#165 · Coronary Artery Disease

In stable ischemic heart disease, which anatomic finding carries a clear survival benefit from revascularization?

Answer

Left main stenosis $\geq 50\%$. For most other stable anatomy, revascularization mainly improves symptoms rather than survival.

#166 · Heart Failure and Cardiomyopathy

Why is a washout required between an ACE inhibitor and sacubitril/valsartan?

Answer

Angioedema risk. Both ACE inhibition and neprilysin inhibition impair bradykinin degradation, and overlap compounds bradykinin accumulation.

#167 · Systemic Disorders Affecting the Circulatory System

Which anticoagulant is preferred for thrombotic antiphospholipid syndrome?

Answer

Warfarin with an INR of 2 to 3. The TRAPS trial showed rivaroxaban increased arterial events in triple-positive patients, so DOACs are avoided.

#168 · Arrhythmias

Which AV blocks are Class 1 pacemaker indications even without symptoms?

Answer

Mobitz type II second-degree block, high-grade AV block, and complete (third-degree) AV block, because the level of block is infranodal and progression is unpredictable.

#169 · Coronary Artery Disease

What is aspirin's current role in primary prevention of ASCVD?

Answer

Low-dose aspirin may be considered in adults 40 to 70 years at higher ASCVD risk without increased bleeding risk. It is not recommended routinely over age 70; secondary prevention use continues indefinitely.

#170 · Arrhythmias

In which two settings is dronedarone contraindicated based on trial evidence of increased mortality?

Answer

Permanent AF (PALLAS) and recent decompensated or advanced (NYHA IV) heart failure (ANDROMEDA). It is a less effective, less toxic amiodarone congener without iodine.

#171 · Coronary Artery Disease

What does the CRUSADE score predict in NSTEMI-ACS?

Answer

In-hospital major bleeding risk, using variables such as hematocrit, creatinine clearance, heart rate, sex, diabetes, and blood pressure. It informs access site and anticoagulant choices.

#172 · Vascular Diseases

What is the immediate first step once acute limb ischemia is suspected?

Answer

Systemic anticoagulation with an IV heparin bolus and infusion, plus emergent vascular specialist consultation, before imaging delays revascularization.

#173 · Arrhythmias

When does sinus node dysfunction justify a permanent pacemaker?

Answer

Only when symptoms clearly correlate with the bradycardia or pauses, or when symptomatic chronotropic incompetence exists. There is no absolute rate or pause number without symptoms.

#174 · Normal Cardiovascular Anatomy and Physiology

Which arteries supply the sinoatrial and atrioventricular nodes?

Answer

The SA nodal artery arises from the right coronary in about 60% of people; the AV nodal artery comes from the dominant artery, the right coronary in roughly 85%.

#175 · Pericardial Disease

How is pericardiocentesis performed for tamponade, and when is it the wrong move?

Answer

Echocardiography-guided drainage is standard; hemopericardium from type A aortic dissection or free-wall rupture needs emergency surgery, not routine needle drainage.

#176 · Arrhythmias

Decode the pacing mode DDD: what do the three letters specify?

Answer

Chamber paced (dual), chamber sensed (dual), response to sensing (dual: triggered and inhibited). A fourth letter R adds rate responsiveness.

#177 · Coronary Artery Disease

Why avoid combining verapamil or diltiazem with a beta blocker, or using them in reduced EF?

Answer

Both drug classes slow AV conduction and depress contractility, risking severe bradycardia, heart block, and decompensated heart failure.

Non-dihydropyridines are avoided when LVEF is reduced.

#178 · Systemic Hypertension and Hypotension

Which four drug classes are first-line for primary hypertension?

Answer

Thiazide or thiazide-like diuretics, ACE inhibitors, angiotensin receptor blockers, and calcium channel blockers.

#179 · Heart Failure and Cardiomyopathy

Name the clinical red flags that should raise suspicion for cardiac amyloidosis.

Answer

LV hypertrophy with low or discordantly normal ECG voltage, bilateral carpal tunnel syndrome, lumbar spinal stenosis, biceps tendon rupture, and intolerance of ACE inhibitors or beta-blockers.

#180 · Pulmonary Circulation Disorders

What hemodynamic triad defines precapillary pulmonary hypertension on right heart catheterization?

Answer

Mean PA pressure >20 mmHg, pulmonary artery wedge pressure ≤ 15 mmHg, and pulmonary vascular resistance >2 Wood units.

#181 · Pulmonary Circulation Disorders

Why can riociguat never be combined with a PDE5 inhibitor?

Answer

Both amplify the nitric oxide/cGMP pathway, and the combination causes severe hypotension. Coadministration is contraindicated, and switching requires a washout period.

#182 · Coronary Artery Disease

In inferior STEMI, ST elevation greater in lead III than lead II suggests which culprit artery?

Answer

The right coronary artery, often with ST depression in lead I or aVL. Circumflex culprits favor elevation in II over III.

#183 · Systemic Hypertension and Hypotension

Which IV agents suit hypertensive emergency with acute pulmonary edema?

Answer

IV nitroglycerin or clevidipine with a loop diuretic. Avoid pure beta blockade in acute decompensated pulmonary edema.

#184 · Arrhythmias

Besides CPVT, which classic toxicity produces bidirectional ventricular tachycardia?

Answer

Digoxin toxicity. Digoxin toxicity also classically causes atrial tachycardia with 2:1 AV block; treat significant toxicity with digoxin immune Fab.

#185 · Heart Failure and Cardiomyopathy

What is the guideline position on MRA, ARNI, and ARB therapy in HFpEF?

Answer

Class 2b options that may reduce hospitalizations, with benefit concentrated at the lower end of the preserved ejection fraction spectrum.

#186 · Heart Failure and Cardiomyopathy

What peak VO₂ on cardiopulmonary exercise testing supports listing for heart transplantation?

Answer

Peak VO₂ ≤14 mL/kg/min, or ≤12 mL/kg/min in patients on a beta-blocker.

#187 · Coronary Artery Disease

How long a nitrate-free interval prevents nitrate tolerance with long-acting therapy?

Answer

10 to 14 hours daily, typically overnight. Continuous 24-hour dosing causes tolerance and loss of antianginal effect.

#188 · Coronary Artery Disease

Outside V2 to V3, what ST elevation qualifies as STEMI on ECG?

Answer

New ST elevation ≥ 1 mm at the J point in at least two contiguous leads.

#189 · Coronary Artery Disease

What is the door-in-door-out goal at a non-PCI hospital transferring a STEMI patient?

Answer

30 minutes or less from arrival to departure for the PCI center.

#190 · Pericardial Disease

What clinical and jugular venous findings suggest constrictive pericarditis at the bedside?

Answer

Elevated jugular venous pressure with a prominent y descent, Kussmaul sign (jugular pressure rising with inspiration), an early diastolic pericardial knock, edema, and ascites.

#191 · Arrhythmias

Does AF pattern (paroxysmal versus persistent) change the decision to anticoagulate for stroke prevention?

Answer

No. Anticoagulation is driven by stroke risk score, not by AF burden or pattern, per the 2023 AF guideline.

#192 · Valvular Disease

Which physical findings point to severe rather than mild aortic stenosis?

Answer

Pulsus parvus et tardus, a late-peaking murmur, and a soft, single, or absent A2 component of S2.

#193 · Arrhythmias

Which CHA2DS2-VASc components score 2 points each, and which score 1 point each?

Answer

Two points: age ≥ 75 and prior stroke, TIA, or thromboembolism. One point each: heart failure, hypertension, diabetes, vascular disease, age 65 to 74, and female sex.

#194 · Heart Failure and Cardiomyopathy

What characterizes anthracycline cardiotoxicity, and at what cumulative doxorubicin dose does risk climb?

Answer

Dose-dependent, largely irreversible myocyte injury; risk rises substantially at cumulative doxorubicin doses ≥ 250 mg/m². Monitor LVEF and global longitudinal strain before and during therapy.

#195 · Valvular Disease

A patient with severe aortic stenosis develops exertional angina, syncope, or heart failure. What is the recommendation?

Answer

Symptomatic severe aortic stenosis is stage D and carries a class 1 indication for aortic valve replacement regardless of ejection fraction.

#196 · Coronary Artery Disease

Name four distinct coronary mechanisms that can produce MINOCA.

Answer

Plaque disruption below angiographic detection, coronary vasospasm, spontaneous coronary artery dissection, and coronary embolism or thrombosis, including microvascular causes.

#197 · Coronary Artery Disease

Spontaneous coronary artery dissection: typical patient, and which systemic condition should be screened for?

Answer

Women under about 50, including peripartum patients, often without atherosclerotic risk factors. Screen for fibromuscular dysplasia with head-to-pelvis vascular imaging.

#198 · Valvular Disease

When should the tricuspid valve be addressed during left-sided valve surgery?

Answer

Severe tricuspid regurgitation at the time of left-sided surgery is class 1; repair is also reasonable for lesser regurgitation with annular dilation above 4.0 cm.

#199 · Valvular Disease

Which findings define uncontrolled infection warranting surgery in endocarditis?

Answer

Paravalvular abscess or fistula, new heart block, and persistent bacteremia or fever 5 to 7 days into appropriate antibiotic therapy.

#200 · Coronary Artery Disease

Fondaparinux in NSTEMI-ACS: dose, renal contraindication, and the PCI-specific hazard?

Answer

2.5 mg subcutaneously daily; contraindicated below a creatinine clearance of 30 mL/min. It must never be the sole anticoagulant during PCI because of catheter thrombosis, so add UFH.

#201 · Valvular Disease

For patients under 50 needing valve replacement, which prosthesis type does the guideline favor?

Answer

A mechanical prosthesis is reasonable if anticoagulation is acceptable, because bioprosthetic valves degenerate substantially faster in younger patients.

#202 · Systemic Hypertension and Hypotension

What is the most common secondary cause of hypertension, and its screening test?

Answer

Primary aldosteronism. Screen with a plasma aldosterone-to-renin ratio; suppressed renin with elevated aldosterone suggests the diagnosis.

#203 · Arrhythmias

AF onset clearly within the last 48 hours in a low stroke-risk patient: can you cardiovert without 3 weeks of anticoagulation?

Answer

Yes, early cardioversion is acceptable; give anticoagulation at presentation and continue at least 4 weeks after. Unstable patients are cardioverted immediately regardless of duration.

#204 · Pulmonary Circulation Disorders

How is acute pulmonary embolism stratified into high, intermediate, and low risk?

Answer

High risk: hemodynamic instability such as arrest, obstructive shock, or persistent hypotension.
Intermediate risk: stable with RV dysfunction or elevated troponin (both defines intermediate-high).
Low risk: none of these.

#205 · Arrhythmias

How is cardioinhibitory carotid sinus hypersensitivity defined, and when does it warrant pacing?

Answer

Carotid sinus massage producing asystole ≥ 3 seconds with reproduction of syncope; pacing is indicated for recurrent syncope from the cardioinhibitory response.

#206 · Heart Failure and Cardiomyopathy

Which natriuretic peptide should be followed in patients taking sacubitril/valsartan, and why?

Answer

NT-proBNP. Neprilysin inhibition raises BNP because BNP is a neprilysin substrate; NT-proBNP is not degraded by neprilysin and remains interpretable.

#207 · Coronary Artery Disease

Ranolazine: mechanism, hemodynamic profile, and its main ECG caution?

Answer

Inhibits the late inward sodium current, reducing diastolic wall tension. It lowers neither heart rate nor blood pressure, but prolongs QT and interacts with strong CYP3A inhibitors.

#208 · Pulmonary Circulation Disorders

State the core principles of managing acute right ventricular failure.

Answer

Avoid excess volume loading, maintain systemic pressure with norepinephrine to preserve RV coronary perfusion, add inotropes for low output, and correct hypoxia and acidosis, which raise pulmonary vascular resistance.

#209 · Coronary Artery Disease

What ECG finding suggests left ventricular aneurysm weeks after anterior MI?

Answer

Persistent ST elevation more than 2 to 4 weeks post-MI, typically in anterior leads, often with deep Q waves. Complications include thrombus, heart failure, and ventricular arrhythmias.

#210 · Pulmonary Circulation Disorders

Name the major PAH drug classes and their molecular targets.

Answer

Endothelin receptor antagonists (ambrisentan, bosentan, macitentan); PDE5 inhibitors (sildenafil, tadalafil); the soluble guanylate cyclase stimulator riociguat; and prostacyclin pathway agents (epoprostenol, treprostinil, iloprost, selexipag).

#211 · Heart Failure and Cardiomyopathy

What distinguishes ACC/AHA Stage C from Stage D heart failure?

Answer

Stage C is structural disease with current or prior symptoms; Stage D is refractory symptoms despite optimized guideline-directed therapy, requiring advanced therapies or palliation.

#212 · Coronary Artery Disease

Which procedural anticoagulant is preferred for PCI in a patient with heparin-induced thrombocytopenia?

Answer

Bivalirudin, a direct thrombin inhibitor that does not cross-react with HIT antibodies. Argatroban is an alternative.

#213 · Valvular Disease

What defines paradoxical low-flow, low-gradient aortic stenosis, and how can severity be confirmed?

Answer

Valve area $\leq 1.0 \text{ cm}^2$, gradient below 40 mmHg, LVEF $\geq 50\%$, and stroke volume index below 35 mL/m^2 ; CT calcium scoring (roughly $\geq 2000 \text{ AU}$ men, $\geq 1200 \text{ AU}$ women) supports true severity.

#214 · Valvular Disease

Name the classic peripheral signs of chronic severe aortic regurgitation.

Answer

Wide pulse pressure with Corrigan water-hammer pulse, de Musset head bobbing, Quincke nail-bed pulsations, Duroziez femoral murmur, and Traube pistol-shot sounds.

#215 · Systemic Disorders Affecting the Circulatory System

When is aspirin started to prevent preeclampsia in high-risk pregnancy?

Answer

Low-dose aspirin (81 mg) begun between 12 and 16 weeks of gestation, before 16 weeks for maximal benefit, and continued until delivery.

#216 · Coronary Artery Disease

Name four absolute contraindications to fibrinolysis in STEMI.

Answer

Any prior intracranial hemorrhage, ischemic stroke within 3 months, active internal bleeding, and suspected aortic dissection. Intracranial malignancy, AVM, and significant recent head trauma also qualify.

#217 · Heart Failure and Cardiomyopathy

Name conditions other than heart failure that raise natriuretic peptide levels.

Answer

Atrial fibrillation, renal failure, older age, pulmonary embolism, pulmonary hypertension, sepsis, and anemia all elevate natriuretic peptides without decompensated heart failure.

#218 · Arrhythmias

What happens when a magnet is placed over a pacemaker versus over an ICD?

Answer

Pacemaker: asynchronous pacing (DOO or VOO) at the magnet rate, ignoring intrinsic activity. ICD: tachyarrhythmia therapies are suspended, but its pacing function is unchanged.

#219 · Systemic Disorders Affecting the Circulatory System

What ejection fraction and timing criteria define peripartum cardiomyopathy?

Answer

New heart failure with LVEF below 45% arising in the last month of pregnancy or within 5 months postpartum, with no other identifiable cause.

#220 · Coronary Artery Disease

After PCI in a patient on oral anticoagulation for AF, how is antithrombotic therapy structured?

Answer

Stop aspirin within 1 to 4 weeks and continue a DOAC plus clopidogrel for 6 to 12 months, then anticoagulant alone. Prolonged triple therapy mainly adds bleeding.

#221 · Arrhythmias

For most patients with AF needing anticoagulation, are DOACs or warfarin preferred, and why?

Answer

DOACs are preferred (Class 1): similar or better stroke prevention with less intracranial hemorrhage than warfarin, except in specific valve situations.

#222 · Heart Failure and Cardiomyopathy

What finding separates SCAI stage B from stage C cardiogenic shock?

Answer

Hypoperfusion. Stage B has relative hypotension without hypoperfusion; stage C shows hypoperfusion, such as elevated lactate, oliguria, cool extremities, or altered mentation, requiring intervention.

#223 · Systemic Disorders Affecting the Circulatory System

How is cancer therapy-related cardiac dysfunction commonly defined on echocardiography?

Answer

An LVEF decline of more than 10 percentage points to a value below 50%, or a relative global longitudinal strain drop over 15% as an early marker.

#224 · Congenital Heart Disease

What hemodynamic findings make ASD closure a Class 1 recommendation in the 2018 ACHD guideline?

Answer

Right atrial or right ventricular enlargement from a significant left-to-right shunt ($Q_p:Q_s$ at least 1.5:1), whether or not symptoms are present, without cyanosis or severe pulmonary hypertension.

#225 · Coronary Artery Disease

How does an LV pseudoaneurysm differ from a true aneurysm, and how is it managed?

Answer

A contained free wall rupture bounded only by pericardium, with a narrow neck relative to its cavity. It requires urgent surgical repair because of high rupture risk.

#226 · Arrhythmias

What ECG triad defines ventricular preexcitation in Wolff-Parkinson-White pattern?

Answer

Short PR interval under 120 ms, a slurred QRS upstroke (delta wave), and a widened QRS from fusion over the accessory pathway.

#227 · Coronary Artery Disease

Cangrelor: route, mechanism, and how fast does platelet function recover after stopping it?

Answer

An intravenous, direct-acting, reversible P2Y₁₂ inhibitor with a 3 to 6 minute half-life. Platelet function recovers within about an hour of stopping the infusion.

#228 · Heart Failure and Cardiomyopathy

How is heart failure with improved ejection fraction (HFimpEF) defined, and does it permit stopping therapy?

Answer

Baseline LVEF $\leq 40\%$, a ≥ 10 point increase, and a second measurement $>40\%$. Continue guideline-directed medical therapy; withdrawal risks relapse (TRED-HF).

#229 · Vascular Diseases

What is first-line drug therapy for Raynaud phenomenon?

Answer

A dihydropyridine calcium channel blocker such as nifedipine or amlodipine, added when cold avoidance and trigger control fail.

#230 · Coronary Artery Disease

List key contraindications to vasodilator stress agents, and the caffeine instruction before testing.

Answer

Active bronchospasm (especially adenosine), high-grade AV block without a pacemaker, and significant hypotension. Hold caffeine and methylxanthines for at least 12 hours.

#231 · Coronary Artery Disease

ST elevation in aVR with diffuse ST depression in multiple leads suggests what anatomy?

Answer

Left main coronary obstruction or severe multivessel disease, especially with hemodynamic compromise. It warrants urgent angiography.

#232 · Arrhythmias

Distinguish pacemaker oversensing from undersensing in terms of pacing behavior.

Answer

Oversensing: noise or T waves are read as beats, so pacing is inappropriately inhibited (pauses).
Undersensing: intrinsic beats are missed, so spikes fire inappropriately after native complexes.

#233 · Heart Failure and Cardiomyopathy

What regional finding in TOPCAT supports spironolactone use in selected HFpEF patients?

Answer

Patients enrolled in the Americas showed reduced heart failure hospitalization with spironolactone, while the Russia/Georgia cohort had implausibly low event rates and undetectable drug metabolites.

#234 · Arrhythmias

A regular narrow-complex tachycardia at exactly 150 bpm should raise suspicion for what rhythm?

Answer

Typical atrial flutter with 2:1 conduction: atrial rate near 300 with sawtooth flutter waves, often unmasked by adenosine or vagal maneuvers.

#235 · Heart Failure and Cardiomyopathy

How is the pulmonary artery pulsatility index (PAPi) calculated, and what value suggests right ventricular failure?

Answer

PAPi = (systolic PA pressure minus diastolic PA pressure) divided by right atrial pressure. Values <0.9 indicate RV failure in acute inferior MI shock.

#236 · Systemic Disorders Affecting the Circulatory System

Describe type 1 amiodarone-induced thyrotoxicosis and its treatment.

Answer

Iodine-driven excess hormone synthesis in a gland with underlying disease, showing increased Doppler vascularity. Treat with methimazole.

#237 · Normal Cardiovascular Anatomy and Physiology

Which leads localize anterior versus lateral infarction, and their arteries?

Answer

V1 to V4 mark the anterior wall from the left anterior descending artery; I, aVL, V5, and V6 mark the lateral wall from the circumflex or diagonal branches.

#238 · Coronary Artery Disease

What is the purpose of the high-sensitivity troponin 0/1-hour algorithm in chest pain evaluation?

Answer

Rapid rule-out or rule-in of NSTEMI: a very low baseline value plus a small 1-hour delta rules out MI with negative predictive value above 99%.

#239 · Pericardial Disease

How does the ECG separate pericarditis from benign early repolarization?

Answer

In V6, an ST-elevation to T-wave amplitude ratio above 0.25 favors pericarditis; early repolarization shows a lower ratio and J-point notching without PR depression.

#240 · Coronary Artery Disease

How long should elective noncardiac surgery be deferred after drug-eluting stent placement?

Answer

Optimally at least 6 months. If surgery cannot wait, 3 months is the minimum, weighing stent thrombosis against surgical delay.

#241 • Coronary Artery Disease

How do you diagnose right ventricular infarction on ECG, and in what setting should you look?

Answer

ST elevation in right-sided lead V4R, checked in every inferior STEMI, especially with hypotension and clear lungs.

#242 • Heart Failure and Cardiomyopathy

What is mavacamten's mechanism, and what LVEF rules govern starting and stopping it?

Answer

A cardiac myosin ATPase inhibitor that reduces LVOT gradients. Start only if LVEF $\geq 55\%$, interrupt if LVEF falls below 50%, and prescribe through a REMS program with attention to CYP interactions.

#243 · Systemic Hypertension and Hypotension

What is the drug rule in pheochromocytoma-related hypertensive crisis?

Answer

Alpha blockade first, with phentolamine IV or oral phenoxybenzamine. Beta blockers before alpha blockade cause unopposed alpha stimulation and worsen the crisis.

#244 · Arrhythmias

What is the first-line drug for torsades de pointes even with a normal serum level?

Answer

IV magnesium sulfate 2 g, regardless of magnesium level. Defibrillate if pulseless or sustained.

#245 · Congenital Heart Disease

What is the classic auscultatory hallmark of an atrial septal defect, and why does it occur?

Answer

Wide, fixed splitting of S2: the shunt equalizes right heart filling across respiration, so P2 timing no longer varies. A pulmonic flow murmur often accompanies it.

#246 · Pulmonary Circulation Disorders

Who should undergo acute vasoreactivity testing in pulmonary hypertension, and what defines a positive response?

Answer

Only idiopathic, heritable, or drug-induced PAH. Positive: mean PA pressure falls ≥ 10 mmHg to an absolute value ≤ 40 mmHg with unchanged or increased cardiac output, usually with inhaled nitric oxide.

#247 · Heart Failure and Cardiomyopathy

How does CRT benefit change with non-LBBB morphology or QRS duration of 120 to 149 ms?

Answer

Benefit weakens and the recommendation drops to class 2a or 2b. Non-LBBB with QRS <150 ms derives the least benefit.

#248 · Systemic Hypertension and Hypotension

What systolic target applies to ambulatory adults aged 65 and older?

Answer

Below 130 mmHg, supported by SPRINT, for community-dwelling older adults. Individualize for frailty, limited life expectancy, or heavy comorbidity.

#249 · Coronary Artery Disease

A DAPT score of what value favors extending dual antiplatelet therapy beyond 12 months?

Answer

≥2 favors prolonged DAPT; a score below 2 favors stopping, because bleeding harm outweighs ischemic benefit.

#250 · Vascular Diseases

What does an ankle-brachial index above 1.40 mean, and what test follows?

Answer

Noncompressible calcified vessels, common in diabetes and CKD, making the ABI unreliable. Obtain a toe-brachial index; below 0.70 diagnoses PAD.

#251 · Vascular Diseases

How does age separate the two major large-vessel vasculitides?

Answer

Takayasu arteritis typically begins before age 40; giant cell arteritis occurs at 50 and older. Both cause granulomatous inflammation of the aorta and its branches.

#252 · Heart Failure and Cardiomyopathy

What is the target dose of sacubitril/valsartan in chronic HFrEF?

Answer

97/103 mg twice daily, uptitrated from 24/26 or 49/51 mg twice daily as tolerated.

#253 · Normal Cardiovascular Anatomy and Physiology

Match physiologic, fixed, and paradoxical S2 splitting to their causes.

Answer

Physiologic splitting widens with inspiration and is normal. Fixed splitting indicates atrial septal defect. Paradoxical splitting, widening on expiration, occurs with LBBB or severe aortic stenosis.

#254 · Systemic Hypertension and Hypotension

When should obstructive sleep apnea be pursued as a cause of hypertension?

Answer

Resistant or nocturnal hypertension with snoring, witnessed apneas, obesity, or daytime sleepiness. Diagnose with polysomnography or home sleep testing.

#255 · Coronary Artery Disease

How soon after starting or changing statin therapy should lipids be rechecked?

Answer

4 to 12 weeks after initiation or dose change, then every 3 to 12 months to confirm adherence and adequate percentage LDL-C reduction.

#256 · Valvular Disease

How is obstructive thrombosis of a left-sided mechanical valve managed?

Answer

Confirm restricted leaflet motion with TEE plus fluoroscopy or CT, then treat with emergency surgery or slow-infusion low-dose fibrinolytic therapy, chosen by thrombus size, symptoms, and surgical risk.

#257 · Systemic Hypertension and Hypotension

In most hypertensive emergencies, how fast should mean arterial pressure fall in the first hour?

Answer

By no more than 25% in the first hour, then toward 160/100 over the next 2 to 6 hours, to preserve organ perfusion.

#258 · Valvular Disease

What is the standard oral endocarditis prophylaxis regimen before a qualifying dental procedure?

Answer

Amoxicillin 2 g orally, given 30 to 60 minutes before the procedure; children receive 50 mg/kg.

#259 · Systemic Disorders Affecting the Circulatory System

What two cardiovascular diseases does chronic methamphetamine use cause?

Answer

A dilated cardiomyopathy that may partially recover with abstinence, and pulmonary arterial hypertension. It also promotes accelerated coronary disease and arrhythmia.

#260 · Coronary Artery Disease

In very high risk ASCVD on maximal statin, what LDL-C level triggers adding nonstatin therapy, and in what order?

Answer

LDL-C ≥ 70 mg/dL. Add ezetimibe first, then a PCSK9 inhibitor if still ≥ 70 mg/dL.

#261 · Systemic Hypertension and Hypotension

What pattern of office and out-of-office readings defines white coat hypertension?

Answer

Elevated office readings with normal out-of-office readings on ambulatory or home monitoring, in a patient not on antihypertensive therapy.

#262 · Normal Cardiovascular Anatomy and Physiology

What valvular events produce the first and second heart sounds?

Answer

S1 is closure of the mitral and tricuspid valves at systole's onset. S2 is closure of the aortic then pulmonic valves at systole's end.

#263 · Coronary Artery Disease

What is the unfractionated heparin bolus for PCI without a glycoprotein IIb/IIIa inhibitor?

Answer

70 to 100 units/kg IV, targeting an activated clotting time of roughly 250 to 300 seconds. With a GP IIb/IIIa inhibitor, use 50 to 70 units/kg.

#264 · Vascular Diseases

How do periprocedural risks differ between carotid stenting and endarterectomy?

Answer

Stenting carries more periprocedural stroke, especially in patients over 70; endarterectomy carries more myocardial infarction and cranial nerve injury. Stenting suits hostile necks and high surgical risk.

#265 · Pericardial Disease

What are the diagnostic criteria for acute pericarditis?

Answer

At least 2 of 4: characteristic pleuritic positional chest pain, pericardial friction rub, new widespread ST elevation or PR depression, and new or worsening pericardial effusion.

#266 · Valvular Disease

How is antithrombotic therapy handled after TAVR in a patient with atrial fibrillation?

Answer

Oral anticoagulation alone; routinely adding an antiplatelet agent increases bleeding without ischemic benefit.

#267 · Heart Failure and Cardiomyopathy

How is cancer therapy-related cardiac dysfunction defined by imaging surveillance?

Answer

An LVEF decline ≥ 10 percentage points to below 50%, or a relative global longitudinal strain reduction $>15\%$ from baseline as an early marker.

#268 · Arrhythmias

In orthodromic AVRT, which direction does the accessory pathway conduct during tachycardia?

Answer

Retrograde. The impulse goes down the AV node and back up the accessory pathway, giving a narrow QRS with a short RP interval and a visible retrograde P after the QRS.

#269 · Arrhythmias

What reentrant circuit produces typical AVNRT, and where does the circuit live?

Answer

Dual AV nodal pathways: slow pathway conducts down, fast pathway conducts back up, entirely within the AV nodal region. It is the most common paroxysmal SVT in adults.

#270 · Heart Failure and Cardiomyopathy

What are the characteristic findings and activity guidance in arrhythmogenic right ventricular cardiomyopathy?

Answer

Desmosomal variants (commonly PKP2), T-wave inversion V1 to V3, epsilon waves, and VT of LBBB morphology from the RV. Restrict high-intensity and competitive exercise, which accelerates disease.

#271 · Heart Failure and Cardiomyopathy

What is the IV to oral conversion for furosemide, and why?

Answer

IV 20 mg is roughly equivalent to oral 40 mg because oral furosemide bioavailability averages about 50%. Torsemide and bumetanide are nearly fully absorbed.

#272 · Arrhythmias

Which DOAC is dosed by three criteria including weight ≤ 60 kg, and which requires twice daily dosing with food?

Answer

Apixaban reduces to 2.5 mg twice daily with 2 of 3: age ≥ 80 , weight ≤ 60 kg, creatinine ≥ 1.5 mg/dL. Rivaroxaban 20 mg is once daily with the evening meal.

#273 · Valvular Disease

What echocardiographic criteria define severe chronic aortic regurgitation?

Answer

Vena contracta above 0.6 cm, effective orifice ≥ 0.3 cm², regurgitant volume ≥ 60 mL, regurgitant fraction $\geq 50\%$, and holodiastolic flow reversal in the descending aorta.

#274 · Valvular Disease

How is a mechanical-valve pregnancy managed in the first trimester when the warfarin dose exceeds 5 mg daily?

Answer

Switch to dose-adjusted twice-daily low molecular weight heparin with anti-Xa peak monitoring, targeting roughly 0.8 to 1.2 U/mL, or use dose-adjusted intravenous heparin.

#275 · Heart Failure and Cardiomyopathy

State the primary prevention ICD criteria for HFrEF with NYHA class II to III symptoms.

Answer

LVEF $\leq 35\%$ despite ≥ 3 months of optimal guideline-directed therapy, with expected meaningful survival greater than 1 year. Applies to ischemic and nonischemic cardiomyopathy.

#276 · Valvular Disease

What did the 2023 Duke-ISCVID update add to endocarditis diagnosis?

Answer

18F-FDG PET/CT and cardiac CT findings as imaging major criteria, intraoperative findings as a new major criterion, and expanded typical organisms and molecular microbiology such as PCR.

#277 · Systemic Hypertension and Hypotension

What heart rate finding points to a neurogenic cause of orthostatic hypotension?

Answer

A blunted compensatory rise, typically under 15 beats per minute despite a large pressure fall, indicating autonomic failure rather than volume depletion.

#278 · Coronary Artery Disease

A coronary artery calcium score of zero in an intermediate-risk adult supports what statin decision?

Answer

Withholding or deferring statin therapy and reassessing in 5 to 10 years, provided the patient does not smoke and lacks diabetes or a strong family history.

#279 · Systemic Disorders Affecting the Circulatory System

When does a GLP-1 receptor agonist carry a cardiovascular indication?

Answer

Type 2 diabetes with established atherosclerotic cardiovascular disease or high risk, where liraglutide and semaglutide reduce major adverse cardiovascular events.

#280 · Pulmonary Circulation Disorders

What hemodynamics distinguish isolated postcapillary from combined pre- and postcapillary pulmonary hypertension?

Answer

Both have wedge pressure >15 mmHg; isolated postcapillary has PVR ≤ 2 Wood units, while combined disease has PVR >2 Wood units.

#281 · Coronary Artery Disease

What is the preferred treatment for post-MI pericarditis, and which drugs are avoided early?

Answer

High-dose aspirin plus colchicine. Avoid NSAIDs and glucocorticoids early after MI because they impair infarct healing and may promote wall thinning.

#282 · Valvular Disease

What is the Carvallo sign and what does it indicate?

Answer

A right-sided murmur, classically tricuspid regurgitation, that grows louder with inspiration because venous return to the right heart increases.

#283 · Coronary Artery Disease

Which patients merit consideration of DAPT extended beyond 12 months, and with what regimen?

Answer

High ischemic risk patients who tolerated a year of DAPT without bleeding, such as prior MI. Ticagrelor 60 mg twice daily plus aspirin is the studied regimen.

#284 · Valvular Disease

Which murmurs get louder with sustained handgrip, and which get softer?

Answer

Handgrip raises afterload, intensifying mitral regurgitation, aortic regurgitation, and VSD murmurs while softening aortic stenosis and HOCM murmurs.

#285 · Valvular Disease

When does embolic risk justify surgery in left-sided endocarditis?

Answer

Recurrent emboli with persistent vegetation despite appropriate antibiotics; surgery is also reasonable for a mobile vegetation longer than 10 mm, particularly on the anterior mitral leaflet.

#286 · Heart Failure and Cardiomyopathy

Name the key contraindications to starting sacubitril/valsartan therapy in heart failure.

Answer

Any history of angioedema, concurrent ACE inhibitor use or use within 36 hours, and pregnancy.

#287 · Systemic Disorders Affecting the Circulatory System

What features suggest Fabry disease in a patient with unexplained LVH?

Answer

X-linked alpha-galactosidase A deficiency mimicking hypertrophic cardiomyopathy, with acroparesthesias, angiokeratomas, anhidrosis, corneal deposits, and proteinuric kidney disease. Enzyme replacement therapy exists, so recognition matters.

#288 · Arrhythmias

How does pacemaker-mediated (endless loop) tachycardia start, and how do devices terminate it?

Answer

A retrogradely conducted atrial beat is sensed and tracked, creating a loop at the upper rate limit. Extending the postventricular atrial refractory period (PVARP) or magnet application breaks it.

#289 · Valvular Disease

What hemodynamic numbers define severe and very severe rheumatic mitral stenosis?

Answer

Severe: valve area ≤ 1.5 cm² with pressure half-time ≥ 150 ms. Very severe: valve area ≤ 1.0 cm² with pressure half-time ≥ 220 ms.

#290 · Coronary Artery Disease

Cardiac rehabilitation after MI, PCI, or CABG: what is its guideline class and typical structure?

Answer

A Class I recommendation. Typically 36 supervised sessions over about 12 weeks, combining exercise training, risk factor management, and education.

#291 · Systemic Disorders Affecting the Circulatory System

How is atrial fibrillation caused by thyrotoxicosis managed acutely and definitively?

Answer

Beta blockade, classically propranolol, for rate control plus definitive treatment of the hyperthyroidism. Many patients revert to sinus rhythm within months of becoming euthyroid.

#292 · Coronary Artery Disease

Is PCI reasonable for STEMI presenting 12 to 24 hours after symptom onset?

Answer

Yes, when there is ongoing ischemia, clinical or electrical instability, or persistent symptoms. Routine PCI of a totally occluded artery beyond 24 hours in a stable asymptomatic patient is not.

#293 · Systemic Hypertension and Hypotension

What is the characteristic side effect of dihydropyridine calcium channel blockers?

Answer

Dose-dependent peripheral edema from precapillary vasodilation. It responds to adding an ACE inhibitor or ARB, not to diuretics.

#294 · Heart Failure and Cardiomyopathy

Why must a monoclonal protein screen precede diagnosing ATTR amyloidosis by bone scintigraphy?

Answer

AL amyloidosis can show myocardial tracer uptake, and missing AL delays urgently needed chemotherapy. Any monoclonal protein requires tissue biopsy with amyloid typing.

#295 · Pericardial Disease

What is first-line anti-inflammatory dosing for acute pericarditis?

Answer

Ibuprofen 600 mg three times daily, or aspirin 750 to 1000 mg three times daily (preferred after myocardial infarction), for 1 to 2 weeks, then tapered.

#296 · Valvular Disease

Which findings trigger intervention in asymptomatic severe aortic stenosis?

Answer

LVEF below 50%, symptoms or a fall in blood pressure on exercise testing, very severe stenosis, or rapid progression; each supports valve replacement.

#297 · Heart Failure and Cardiomyopathy

Which patient profile gains the strongest class 1 benefit from cardiac resynchronization therapy?

Answer

LVEF $\leq 35\%$, sinus rhythm, left bundle branch block with QRS ≥ 150 ms, and NYHA class II to IV symptoms on guideline-directed therapy.

#298 · Coronary Artery Disease

How does morphine interact with oral P2Y12 inhibitors in ACS?

Answer

Morphine delays gastric emptying, slowing P2Y12 inhibitor absorption and delaying platelet inhibition. Use it only for unacceptable pain despite anti-ischemic therapy.

#299 · Systemic Disorders Affecting the Circulatory System

What cardiac and systemic clues suggest hemochromatosis, and the screening tests?

Answer

Cardiomyopathy, often restrictive early then dilated, with conduction disease, plus diabetes, bronze skin, arthropathy, and liver disease. Screen with ferritin and transferrin saturation.

#300 · Heart Failure and Cardiomyopathy

Why are thiazolidinediones such as pioglitazone avoided in symptomatic heart failure?

Answer

They cause fluid retention and increase heart failure hospitalization, so they are harmful in NYHA class II to IV heart failure.

#301 · Valvular Disease

Describe the auscultatory findings of mitral stenosis and the marker of worsening severity.

Answer

Loud S1, an opening snap after S2, and a low-pitched apical diastolic rumble in left lateral decubitus; a shorter A2-to-opening-snap interval indicates more severe stenosis.

#302 · Pericardial Disease

Which ECG features favor pericarditis over ST-elevation myocardial infarction?

Answer

Diffuse concave ST elevation without reciprocal depression (except aVR), PR depression, and no Q waves or territorial pattern; STEMI shows convex, regional elevation with reciprocal changes.

#303 • Coronary Artery Disease

LV thrombus after MI: which infarct location predisposes, and what anticoagulation is standard?

Answer

Anterior MI with apical akinesis or aneurysm. Treat with warfarin for about 3 months, with imaging to confirm resolution; DOACs are an increasingly used alternative.

#304 • Pericardial Disease

What bedside findings define cardiac tamponade, including the pulsus paradoxus threshold?

Answer

Hypotension, elevated jugular venous pressure, and muffled heart sounds (Beck triad), with an inspiratory systolic pressure fall greater than 10 mmHg defining pulsus paradoxus.

#305 • Coronary Artery Disease

Why is simvastatin 80 mg no longer initiated, and what interactions limit lower doses?

Answer

The 80 mg dose carries excess myopathy and rhabdomyolysis risk, so 40 mg is the practical maximum. Strong CYP3A4 inhibitors and gemfibrozil further raise myopathy risk.

#306 • Coronary Artery Disease

What findings 60 to 90 minutes after fibrinolysis indicate failed reperfusion and the need for rescue PCI?

Answer

Less than 50% resolution of the maximal ST elevation, persistent chest pain, or hemodynamic or electrical instability.

#307 · Congenital Heart Disease

Name three clinical or radiographic clues to undiagnosed aortic coarctation in an adult.

Answer

Upper extremity hypertension with radial-femoral pulse delay, a systolic murmur over the back, rib notching and the figure 3 sign on chest radiograph, and diminished lower extremity pressures.

#308 · Systemic Hypertension and Hypotension

Which antihypertensive drug classes are contraindicated during pregnancy, and why?

Answer

ACE inhibitors, ARBs, and direct renin inhibitors, which cause fetal renal injury, oligohydramnios, and skeletal defects. Mineralocorticoid receptor antagonists are also avoided.

#309 · Arrhythmias

List four commonly tested drug classes that prolong the QT and can precipitate torsades.

Answer

Class IA and III antiarrhythmics (quinidine, sotalol, dofetilide, ibutilide), macrolides and fluoroquinolones, azole antifungals, antipsychotics such as haloperidol, and methadone. Hypokalemia and hypomagnesemia compound the risk.

#310 · Arrhythmias

What is the definitive treatment for symptomatic WPW with documented tachyarrhythmia?

Answer

Catheter ablation of the accessory pathway, with success rates above 95% and low complication rates.

#311 · Valvular Disease

What quantitative criteria define severe primary mitral regurgitation?

Answer

Effective regurgitant orifice ≥ 0.40 cm², regurgitant volume ≥ 60 mL, regurgitant fraction $\geq 50\%$, and vena contracta ≥ 0.7 cm.

#312 · Valvular Disease

What echocardiographic numbers define severe high-gradient aortic stenosis?

Answer

Peak aortic velocity ≥ 4 m/s or mean gradient ≥ 40 mmHg, with valve area typically ≤ 1.0 cm².

#313 · Coronary Artery Disease

Within what symptom-onset window is fibrinolysis indicated for STEMI when timely PCI is unavailable?

Answer

Within 12 hours of symptom onset. Benefit is greatest in the first 3 hours and falls steadily with delay.

#314 · Valvular Disease

A pregnant patient with a mechanical valve takes warfarin 4 mg daily. What is the first-trimester strategy?

Answer

Continuing warfarin is reasonable when the dose is ≤ 5 mg daily, since it carries the lowest maternal valve thrombosis risk with low embryopathy risk at that dose.

#315 · Systemic Hypertension and Hypotension

What is midodrine's mechanism and its key dosing precaution?

Answer

An alpha-1 agonist prodrug that raises standing pressure. Do not dose within 4 hours of bedtime because it causes supine hypertension.

#316 · Heart Failure and Cardiomyopathy

What is dobutamine's mechanism, and what chronic therapy blunts its effect?

Answer

A beta-1 receptor agonist increasing contractility. Chronic beta-blocker therapy competitively blunts its effect, so higher doses or a different agent may be needed.

#317 · Congenital Heart Disease

What is the most common ASD type, and which type is amenable to percutaneous device closure?

Answer

Secundum ASD, in the fossa ovalis, is most common and the only type routinely closed percutaneously, given adequate rims.

#318 · Congenital Heart Disease

Explain the VSD murmur paradox: why does a small defect sound louder than a large one?

Answer

A small restrictive VSD maintains a high interventricular pressure gradient, producing a loud harsh holosystolic murmur; a large nonrestrictive defect equalizes pressures and can be nearly silent.

#319 · Heart Failure and Cardiomyopathy

What is the genetic basis and inheritance pattern of hypertrophic cardiomyopathy?

Answer

Autosomal dominant sarcomere gene variants, most commonly MYBPC3 and MYH7. First-degree relatives need clinical screening, with cascade genetic testing when a pathogenic variant is known.

#320 · Coronary Artery Disease

What dose of colchicine may be considered for secondary prevention in chronic coronary disease, and why?

Answer

0.5 mg daily, an anti-inflammatory strategy shown in LoDoCo2 and COLCOT to reduce recurrent cardiovascular events.

#321 · Heart Failure and Cardiomyopathy

What NT-proBNP level effectively rules out acute heart failure in the emergency setting?

Answer

NT-proBNP <300 pg/mL has a high negative predictive value for excluding acute heart failure.

#322 · Heart Failure and Cardiomyopathy

What counseling applies to future pregnancy after peripartum cardiomyopathy?

Answer

Subsequent pregnancy carries substantial risk of relapse and death when LVEF has not normalized, so it is discouraged with persistent LV dysfunction.

#323 · Congenital Heart Disease

Which valve lesion and ECG finding accompany an ostium primum ASD?

Answer

A cleft anterior mitral leaflet with mitral regurgitation, plus left axis deviation with first-degree AV block on ECG. It is part of the AV septal (endocardial cushion) defect spectrum.

#324 · Heart Failure and Cardiomyopathy

How is the warm and wet acute heart failure profile managed?

Answer

Congested but well perfused, the most common presentation: treat with IV loop diuretics, adding vasodilators for hypertension or refractory congestion. Inotropes are not indicated.

#325 · Coronary Artery Disease

After successful fibrinolysis for STEMI, when should transfer for angiography occur?

Answer

Routinely within 3 to 24 hours of lysis, the pharmacoinvasive strategy, even when reperfusion appears successful.

#326 · Systemic Hypertension and Hypotension

What systolic and diastolic readings define stage 1 hypertension?

Answer

Systolic 130 to 139 mmHg or diastolic 80 to 89 mmHg.

#327 · Valvular Disease

At what aortic diameter does a bicuspid aortic valve patient need aortic surgery?

Answer

At ≥ 5.5 cm; at ≥ 5.0 cm it is reasonable with risk factors such as family history of dissection or growth ≥ 0.5 cm per year, or at low risk in an experienced center.

#328 · Valvular Disease

What happens to the HOCM murmur and the mitral valve prolapse click with squatting versus standing?

Answer

Squatting raises preload and afterload, softening HOCM and moving the MVP click later; standing does the opposite, making both earlier and louder.

#329 · Systemic Hypertension and Hypotension

What blood pressure is required before giving thrombolytics in acute ischemic stroke?

Answer

Below 185/110 mmHg before the lytic, then maintained below 180/105 mmHg for 24 hours afterward.

#330 · Coronary Artery Disease

How can DAPT be abbreviated after PCI in patients at high bleeding risk?

Answer

Stop one agent after 1 to 3 months and continue P2Y12 inhibitor monotherapy, an approach shown to cut bleeding without an ischemic penalty.

#331 · Valvular Disease

How often should asymptomatic aortic stenosis be re-imaged by severity?

Answer

Every 3 to 5 years for mild disease, every 1 to 2 years for moderate, and every 6 to 12 months for severe stenosis.

#332 · Heart Failure and Cardiomyopathy

What is first-line medical therapy for symptomatic obstructive hypertrophic cardiomyopathy?

Answer

A nonvasodilating beta-blocker. Verapamil or diltiazem is the alternative if beta-blockers are ineffective or not tolerated.

#333 · Valvular Disease

What is the first-line treatment for severe secondary mitral regurgitation with reduced ejection fraction?

Answer

Optimized guideline-directed medical therapy for heart failure, plus cardiac resynchronization when indicated, before any valve intervention is considered.

#334 · Coronary Artery Disease

In patients ≥ 75 years receiving a pharmacoinvasive strategy, how is tenecteplase dosing adjusted?

Answer

Give half the standard weight-based dose.
Full-dose tenecteplase in the elderly caused excess intracranial hemorrhage in the STREAM trial.

#335 · Heart Failure and Cardiomyopathy

What distinguishes ACC/AHA Stage A from Stage B heart failure?

Answer

Stage A is at risk with no structural disease or symptoms; Stage B is pre-heart failure with structural disease, elevated filling pressures, or raised natriuretic peptides but no symptoms.

#336 · Arrhythmias

What is the initial adult IV adenosine dose for SVT, and how must it be given?

Answer

6 mg by rapid IV push with an immediate saline flush; if ineffective, give 12 mg. Its half-life is only seconds.

#337 · Heart Failure and Cardiomyopathy

Which three beta-blockers have mortality evidence in HFrEF and are guideline endorsed?

Answer

Carvedilol, metoprolol succinate, and bisoprolol. Other beta-blockers lack comparable HFrEF mortality evidence and should not substitute.

#338 · Normal Cardiovascular Anatomy and Physiology

Give the normal adult ranges for the PR, QRS, and QTc intervals.

Answer

PR interval 120 to 200 ms, QRS duration below 120 ms, and corrected QT below 450 ms in men and 460 ms in women.

#339 · Normal Cardiovascular Anatomy and Physiology

State the Fick principle for measuring cardiac output.

Answer

Cardiac output equals oxygen consumption divided by the arteriovenous oxygen content difference: $CO = VO_2 \div (\text{arterial } O_2 \text{ content} - \text{venous } O_2 \text{ content})$.

#340 · Coronary Artery Disease

Wellens syndrome: what is the ECG pattern and what lesion does it signal?

Answer

Biphasic or deeply inverted T waves in V2 to V3 during a pain-free interval, signaling critical proximal LAD stenosis and high risk of anterior MI.

#341 · Vascular Diseases

What is the first-line therapy for intermittent claudication, and its prescription?

Answer

Supervised exercise: walking to near-maximal claudication pain, 30 to 45 minutes per session, at least 3 sessions weekly for at least 12 weeks.

#342 · Pericardial Disease

What is Dressler syndrome and when does it occur after myocardial infarction?

Answer

An immune-mediated pericarditis with fever and effusion appearing weeks to months after infarction, typically 1 to 8 weeks, treated with aspirin-based therapy plus colchicine.

#343 · Vascular Diseases

Why does Loeys-Dietz syndrome warrant aortic repair at smaller diameters?

Answer

Its aortas dissect at smaller sizes than Marfan, so repair is considered around 4.5 cm depending on genotype and family history.

#344 · Valvular Disease

Which procedures warrant endocarditis prophylaxis in qualifying patients?

Answer

Dental procedures manipulating gingival tissue or the periapical region, or perforating oral mucosa; routine GI or GU procedures do not qualify without active infection.

#345 · Heart Failure and Cardiomyopathy

Why is ARNI preferred over an ACE inhibitor in symptomatic chronic HFrEF?

Answer

Sacubitril/valsartan reduced cardiovascular death and heart failure hospitalization by 20% versus enalapril in PARADIGM-HF, earning a class 1 preference.

#346 · Heart Failure and Cardiomyopathy

Name the four pillars of guideline-directed medical therapy for HFrEF.

Answer

ARNI (or ACEi/ARB), an evidence-based beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT2 inhibitor.

#347 · Systemic Hypertension and Hypotension

What is the biochemical screening test of choice for pheochromocytoma?

Answer

Plasma free metanephrines or 24-hour urine fractionated metanephrines, both highly sensitive. Confirm imaging only after biochemical diagnosis.

#348 · Valvular Disease

Endocarditis grows *Streptococcus gallolyticus* (formerly *bovis*). What workup is mandatory?

Answer

Colonoscopy, because of the strong association with colorectal neoplasia.

#349 · Heart Failure and Cardiomyopathy

What disease-modifying drug is approved for transthyretin amyloid cardiomyopathy, and how does it work?

Answer

Tafamidis, a transthyretin stabilizer that reduced all-cause mortality and cardiovascular hospitalization in ATTR-ACT.

#350 · Valvular Disease

Which organism classically causes subacute native valve endocarditis after dental procedures?

Answer

Viridans group streptococci, oral flora that seed previously damaged valves during gingival bacteremia.

#351 · Coronary Artery Disease

What outcome benefits does cardiac rehabilitation deliver after MI?

Answer

Reduced cardiovascular mortality, fewer hospital readmissions, and improved exercise capacity and quality of life. It remains substantially underused, especially in women and older adults.

#352 · Vascular Diseases

At what diameters is elective abdominal aortic aneurysm repair indicated?

Answer

≥5.5 cm in men and ≥5.0 cm in women, or growth over 0.5 cm in 6 months, or any symptomatic aneurysm.

#353 · Coronary Artery Disease

How long must you wait after MI, and after revascularization, before implanting a primary prevention ICD?

Answer

At least 40 days after MI and 90 days after revascularization, with persisting LVEF $\leq 35\%$ despite guideline-directed medical therapy.

#354 · Coronary Artery Disease

How many days before elective CABG should prasugrel be held?

Answer

At least 7 days, the longest hold of the oral P2Y₁₂ inhibitors, reflecting its potent irreversible inhibition.

#355 • Valvular Disease

What findings on transesophageal echo contraindicate balloon mitral commissurotomy?

Answer

Left atrial thrombus or more than mild mitral regurgitation; TEE must exclude both before the procedure.

#356 • Heart Failure and Cardiomyopathy

How is peripartum cardiomyopathy defined, and which HF drugs are unsafe before delivery?

Answer

LVEF <45% developing toward the end of pregnancy or in the months after delivery without another cause. ACE inhibitors, ARBs, ARNI, and MRAs are contraindicated during pregnancy.

#357 · Valvular Disease

How does Valsalva strain separate the murmur of hypertrophic obstructive cardiomyopathy from valvular aortic stenosis?

Answer

Valsalva decreases preload, so the HOCM murmur grows louder while the aortic stenosis murmur softens.

#358 · Pulmonary Circulation Disorders

Which agent is preferred for high-risk PAH with right heart failure, and what is its key practical limitation?

Answer

Intravenous epoprostenol, the agent with mortality benefit. Its half-life is only a few minutes, so abrupt infusion interruption can cause life-threatening rebound pulmonary hypertension.

#359 · Coronary Artery Disease

Name the two PCSK9 monoclonal antibodies and their approximate LDL-C reduction.

Answer

Evolocumab and alirocumab, injected every 2 to 4 weeks, each lowering LDL-C by about 60%.

#360 · Arrhythmias

Long QT type 3: when do events occur, what is the channel defect, and which adjunct drug helps?

Answer

Events occur at rest or during sleep; the defect is SCN5A sodium channel gain of function. Mexiletine, a sodium channel blocker, can shorten the QT as an adjunct.

#361 · Valvular Disease

When does chronic severe primary mitral regurgitation warrant surgery?

Answer

With symptoms, or in asymptomatic patients once LVEF is $\leq 60\%$ or LVESD reaches ≥ 40 mm; both situations are class 1.

#362 · Valvular Disease

Describe the aortic regurgitation murmur and how to bring it out on exam.

Answer

A high-pitched blowing decrescendo diastolic murmur at the left sternal border, loudest with the patient sitting, leaning forward, in end-expiration; handgrip intensifies it.

#363 · Coronary Artery Disease

Bempedoic acid: mechanism, why it spares muscle, and two notable adverse effects?

Answer

Inhibits ATP-citrate lyase upstream of HMG-CoA reductase and is activated only in the liver, so myalgia is rare. It raises uric acid, causing gout, and increases tendon rupture risk.

#364 · Valvular Disease

What is the dimensionless index, and what value suggests severe aortic stenosis?

Answer

The ratio of LVOT to aortic valve velocity-time integrals; a value below 0.25 suggests severe stenosis and is relatively independent of flow.

#365 · Arrhythmias

What causes pacemaker syndrome, and what is the fix?

Answer

Loss of AV synchrony with single-chamber ventricular (VVI) pacing: atria contract against closed AV valves, causing fatigue, fullness, and hypotension. Upgrade to dual-chamber (atrial-tracking) pacing.

#366 · Congenital Heart Disease

In congenitally corrected transposition (L-TGA), which ventricle supports the systemic circulation, and what conduction problem accrues?

Answer

The morphologic right ventricle, which fails over decades with systemic AV valve regurgitation. Spontaneous complete AV block develops at roughly 2% per year.

#367 · Heart Failure and Cardiomyopathy

What wall thickness defines hypertrophic cardiomyopathy in an adult without another explanation?

Answer

Maximal LV wall thickness ≥ 15 mm not explained by loading conditions; ≥ 13 mm suffices with a positive family history or pathogenic sarcomere variant.

#368 · Coronary Artery Disease

State the four 10-year ASCVD risk bands from the Pooled Cohort Equations.

Answer

Low under 5%, borderline 5 to under 7.5%, intermediate 7.5 to under 20%, and high $\geq 20\%$.
Statin discussion begins at intermediate risk.

#369 · Coronary Artery Disease

Why are nitrates avoided in right ventricular infarction?

Answer

The failing RV is preload dependent, so venodilation can cause profound hypotension. Treat hypotension with IV fluids and reperfusion instead.

#370 · Pulmonary Circulation Disorders

What is the screening test of choice for chronic thromboembolic pulmonary hypertension (CTEPH)?

Answer

Ventilation/perfusion scintigraphy. A normal V/Q scan effectively excludes CTEPH, outperforming CT pulmonary angiography for screening.

#371 · Vascular Diseases

What clinical triad suggests a diagnosis of chronic mesenteric ischemia?

Answer

Postprandial abdominal pain within an hour of eating, fear of food, and unintentional weight loss, usually with multivessel mesenteric stenosis.

#372 · Heart Failure and Cardiomyopathy

Which drug classes worsen outflow obstruction in obstructive HCM and should be avoided?

Answer

Pure vasodilators such as dihydropyridines and nitrates, and aggressive diuresis. Both reduce preload or afterload and increase the dynamic gradient.

#373 · Vascular Diseases

What Rutherford acute limb ischemia finding mandates emergent revascularization?

Answer

Class IIb: motor weakness with sensory loss beyond the toes, an inaudible arterial Doppler signal, but an audible venous signal. The limb is immediately threatened and needs revascularization within hours.

#374 · Systemic Disorders Affecting the Circulatory System

What are cocaine's main cardiac effects and the initial chest pain management?

Answer

Coronary vasospasm, thrombosis, tachycardia, and hypertension causing ischemia. Treat with benzodiazepines, nitroglycerin, and aspirin; avoid beta blockers in the acute intoxicated phase.

#375 · Coronary Artery Disease

Contrast the timing of peri-infarction pericarditis with Dressler syndrome.

Answer

Peri-infarction pericarditis occurs within 1 to 4 days of transmural MI. Dressler syndrome is immune mediated, appearing weeks to months later, typically 2 to 8 weeks.

#376 · Valvular Disease

What conditions accompany bicuspid aortic valve, and who else should be screened?

Answer

Aortopathy of the root and ascending aorta and aortic coarctation; screening echocardiography of first-degree relatives is reasonable.

#377 · Valvular Disease

What antithrombotic therapy follows surgical bioprosthetic valve replacement?

Answer

Aspirin 75 to 100 mg daily; a vitamin K antagonist to INR 2.5 for 3 to 6 months is also reasonable when bleeding risk is low.

#378 · Heart Failure and Cardiomyopathy

Why should NSAIDs be avoided in patients with heart failure?

Answer

They promote sodium and water retention, blunt diuretic and RAAS inhibitor effect, worsen renal function, and increase heart failure hospitalization risk.

#379 · Systemic Disorders Affecting the Circulatory System

Describe type 2 amiodarone-induced thyrotoxicosis and its treatment.

Answer

A destructive thyroiditis releasing preformed hormone from a normal gland, with low uptake and low Doppler vascularity. Treat with glucocorticoids.

#380 · Heart Failure and Cardiomyopathy

How do ACC/AHA heart failure stages differ conceptually from NYHA functional classes?

Answer

Stages describe disease progression and only advance forward; NYHA class grades current symptom severity and can move in both directions with treatment.

#381 · Systemic Hypertension and Hypotension

Name clinical triggers that should prompt a secondary hypertension workup.

Answer

Onset before age 30, resistant hypertension, abrupt worsening of controlled pressure, unprovoked hypokalemia, severe or malignant hypertension, and disproportionate target organ damage.

#382 · Systemic Disorders Affecting the Circulatory System

At what cumulative doxorubicin dose does cardiotoxicity risk rise substantially?

Answer

Risk climbs with cumulative dose and becomes substantial above roughly 250 mg/m². Toxicity is dose-dependent and often irreversible, so obtain baseline and surveillance echocardiography.

#383 · Vascular Diseases

When is revascularization considered for asymptomatic carotid stenosis?

Answer

Selected patients with $\geq 70\%$ stenosis, low periprocedural risk, and life expectancy over 5 years, alongside intensive medical therapy. Medical therapy alone is reasonable for many.

#384 · Arrhythmias

What surveillance testing does chronic amiodarone therapy require, and at what interval?

Answer

Thyroid and liver function tests at baseline and roughly every 6 months, a baseline chest radiograph, and pulmonary evaluation (PFTs with DLCO) for new respiratory symptoms; periodic eye and skin review.

#385 · Valvular Disease

Name key causes of culture-negative endocarditis and how they are identified.

Answer

Coxiella burnetii, *Bartonella* species, *Brucella*, and *Tropheryma whippelii*; diagnosis relies on serology, PCR, or valve tissue testing rather than routine cultures.

#386 · Valvular Disease

Which COAPT-based criteria support transcatheter edge-to-edge repair for secondary mitral regurgitation?

Answer

Persistent symptoms on optimal medical therapy, suitable anatomy, LVEF 20 to 50%, LVESD ≤ 70 mm, and pulmonary artery systolic pressure ≤ 70 mmHg.

#387 · Arrhythmias

What arrhythmia pattern, trigger, and gene define catecholaminergic polymorphic ventricular tachycardia (CPVT)?

Answer

Bidirectional or polymorphic VT with exercise or emotion, a normal resting ECG, and RyR2 ryanodine receptor mutations. Treat with nonselective beta blockers, adding flecainide if needed.

#388 · Heart Failure and Cardiomyopathy

Which SGLT2 inhibitors and doses are guideline endorsed for HFrEF regardless of diabetes status?

Answer

Dapagliflozin 10 mg daily and empagliflozin 10 mg daily, each with benefit independent of diabetes status (DAPA-HF, EMPEROR-Reduced).

#389 · Systemic Hypertension and Hypotension

What 24-hour, daytime, and home blood pressure means correspond to an office reading of 130/80 mmHg?

Answer

24-hour mean $\geq 125/75$, daytime mean $\geq 130/80$, and home average $\geq 130/80$ mmHg.

#390 · Coronary Artery Disease

What is the first-line antianginal class in chronic stable angina, and the usual resting heart rate goal?

Answer

Beta blockers, titrated to a resting heart rate of about 55 to 60 beats per minute.

#391 · Coronary Artery Disease

What arrhythmia causes most pre-hospital deaths in acute MI, and what is its late prognostic meaning?

Answer

Ventricular fibrillation. Within the first 48 hours it does not by itself mandate an ICD, but VF or sustained VT beyond 48 hours signals high arrhythmic risk.

#392 · Heart Failure and Cardiomyopathy

What is the target dose range of spironolactone in HFrEF?

Answer

25 to 50 mg once daily, typically started at 12.5 to 25 mg daily.

#393 · Coronary Artery Disease

Vasospastic (Prinzmetal) angina: first-line therapy and which drug class can worsen it?

Answer

Calcium channel blockers, often with long-acting nitrates. Nonselective beta blockers can worsen spasm by leaving alpha-mediated constriction unopposed.

#394 · Arrhythmias

Why must an AV nodal blocker accompany flecainide when treating atrial fibrillation or flutter?

Answer

Class IC drugs can organize AF into slow atrial flutter that conducts 1:1, paradoxically accelerating the ventricular rate. A beta blocker or nondihydropyridine calcium blocker prevents this.

#395 · Arrhythmias

For ischemic cardiomyopathy with NYHA class I symptoms, what EF threshold earns a primary prevention ICD?

Answer

LVEF $\leq 30\%$, again at least 40 days post MI and 90 days post revascularization on optimal medical therapy.

#396 · Congenital Heart Disease

When is ASD closure contraindicated because of pulmonary vascular disease?

Answer

When PA systolic pressure exceeds two thirds of systemic, PVR exceeds two thirds of systemic resistance, or there is a net right-to-left shunt (Eisenmenger physiology). Closure would remove the pop-off.

#397 · Vascular Diseases

Which symptomatic carotid stenosis benefits most from endarterectomy, and by when?

Answer

70 to 99% stenosis after a nondisabling stroke or TIA, ideally within 2 weeks of the event. Men with 50 to 69% stenosis gain more modest benefit.

#398 · Pulmonary Circulation Disorders

Name the five WHO clinical groups of pulmonary hypertension.

Answer

Group 1 pulmonary arterial hypertension; group 2 left heart disease; group 3 lung disease or hypoxia; group 4 pulmonary artery obstruction including CTEPH; group 5 unclear or multifactorial mechanisms.

#399 · Valvular Disease

What prophylaxis options exist for penicillin-allergic patients, and which drug was removed?

Answer

Cephalexin 2 g, azithromycin or clarithromycin 500 mg, or doxycycline 100 mg; the 2021 AHA update removed clindamycin because of adverse events including *C. difficile*.

#400 · Congenital Heart Disease

Which ECG measurement predicts ventricular tachycardia and sudden death in repaired tetralogy of Fallot?

Answer

QRS duration ≥ 180 ms, reflecting RV dilation and slowed conduction; it flags patients needing closer arrhythmia surveillance and evaluation.

#401 · Vascular Diseases

Contrast atherosclerotic renal artery stenosis with fibromuscular dysplasia.

Answer

Atherosclerosis affects the ostium and proximal artery in older patients. Fibromuscular dysplasia causes a mid-to-distal string-of-beads appearance in young women and responds to balloon angioplasty without stenting.

#402 · Valvular Disease

How is atrial fibrillation anticoagulated in rheumatic mitral stenosis?

Answer

With warfarin regardless of CHA₂DS₂-VASc score; direct oral anticoagulants are not recommended in rheumatic mitral stenosis or with mechanical valves.

#403 · Vascular Diseases

What defines a complicated type B dissection requiring intervention?

Answer

Malperfusion of viscera, kidneys, or limbs, rupture or impending rupture, refractory pain, refractory hypertension, or rapid aortic expansion.

#404 · Coronary Artery Disease

Which post-MI patients should receive an ACE inhibitor within 24 hours?

Answer

Those with anterior infarction, heart failure, or LVEF $\leq 40\%$, provided blood pressure allows. Use an ARB if ACE inhibitor intolerant.

#405 · Heart Failure and Cardiomyopathy

What EF threshold qualifies an asymptomatic NYHA class I ischemic cardiomyopathy patient for a primary prevention ICD?

Answer

LVEF $\leq 30\%$ at least 40 days post-MI on guideline-directed therapy, with expected survival greater than 1 year.

#406 · Valvular Disease

In the 2020 ACC/AHA framework, which aortic stenosis patients should preferentially receive transfemoral TAVR by age?

Answer

Patients over 80, or with life expectancy under 10 years, are preferentially treated with transfemoral TAVR when anatomy is suitable.

#407 · Heart Failure and Cardiomyopathy

How long must you wait after myocardial infarction and after revascularization before implanting a primary prevention ICD?

Answer

At least 40 days after MI and at least 90 days after revascularization, allowing time for LVEF recovery.

#408 · Coronary Artery Disease

STEMI requiring transfer to a PCI center: what total first medical contact to device time still favors primary PCI?

Answer

120 minutes or less. If PCI cannot be achieved within 120 minutes, give fibrinolysis unless contraindicated.

#409 · Coronary Artery Disease

How is unstable angina defined now that high-sensitivity troponin is standard?

Answer

Ischemic symptoms at rest or with minimal exertion, new or accelerating, without troponin elevation above the 99th percentile. It is increasingly rare with high-sensitivity assays.

#410 · Systemic Disorders Affecting the Circulatory System

What presentation should trigger evaluation for cardiac sarcoidosis?

Answer

Unexplained high-grade AV block in a patient younger than 60, or unexplained ventricular tachycardia. Evaluate with cardiac MRI and FDG-PET.

#411 · Heart Failure and Cardiomyopathy

Why is milrinone effective in patients on chronic beta-blockade, and what are its main cautions?

Answer

As a phosphodiesterase-3 inhibitor it acts downstream of the beta receptor. It vasodilates, so avoid it in hypotension, and it accumulates in renal failure because of renal clearance.

#412 · Arrhythmias

At what CHA₂DS₂-VASc score does the 2023 AF guideline recommend anticoagulation for men and for women?

Answer

Score ≥ 2 in men or ≥ 3 in women (annual stroke risk $\geq 2\%$). A score of 1 in men or 2 in women makes anticoagulation reasonable.

#413 · Valvular Disease

What antithrombotic regimen follows TAVR in a patient without another anticoagulation indication?

Answer

Aspirin 75 to 100 mg daily alone; single antiplatelet therapy caused less bleeding than dual therapy without more ischemic events in POPular TAVI.

#414 · Heart Failure and Cardiomyopathy

What does the I-NEED-HELP mnemonic identify, and what do its letters stand for?

Answer

Triggers for advanced heart failure referral: IV inotropes, NYHA IIIB/IV or rising natriuretic peptides, End-organ dysfunction, EF $\leq$ 35%, Defibrillator shocks, Hospitalizations >1, Edema despite escalating diuretics, Low blood pressure, Prognostic medication intolerance.

#415 · Heart Failure and Cardiomyopathy

How is immune checkpoint inhibitor myocarditis recognized and treated?

Answer

Troponin elevation, arrhythmias, or new LV dysfunction on checkpoint inhibitor therapy, sometimes with concurrent myositis and myasthenia gravis. Hold the checkpoint inhibitor and give high-dose corticosteroids promptly.

#416 · Heart Failure and Cardiomyopathy

Is a washout period needed when switching from an ARB to sacubitril/valsartan?

Answer

No. ARBs do not affect bradykinin degradation, so the next scheduled dose can simply be replaced by the ARNI.

#417 · Congenital Heart Disease

Which congenital heart patients still warrant endocarditis prophylaxis before dental procedures?

Answer

Unrepaired cyanotic CHD including palliative shunts, repairs with prosthetic material within the first 6 months, residual defects adjacent to prosthetic material, and prior endocarditis or prosthetic valves.

#418 · Pericardial Disease

How long is colchicine continued for recurrent pericarditis?

Answer

Six months, twice the duration used for a first episode, alongside the anti-inflammatory taper.

#419 · Coronary Artery Disease

After CABG performed for ACS, how long should DAPT continue?

Answer

12 months, the same as any ACS, then aspirin monotherapy indefinitely.

#420 · Systemic Hypertension and Hypotension

How is hypertensive urgency defined, and how should it be managed?

Answer

Blood pressure $\geq 180/120$ mmHg without acute target organ damage. Manage with oral agents and gradual reduction over 24 to 48 hours, not IV therapy or ICU admission.

#421 · Arrhythmias

After any cardioversion of AF, how long must anticoagulation continue regardless of stroke risk score?

Answer

At least 4 weeks, because atrial mechanical stunning persists after electrical or pharmacologic conversion. Continue longer if CHA₂DS₂-VASc warrants.

#422 · Systemic Disorders Affecting the Circulatory System

Summarize the normal hemodynamic changes that occur during a healthy pregnancy.

Answer

Plasma volume rises 40 to 50%, cardiac output rises 30 to 50%, and systemic vascular resistance falls, with blood pressure reaching its nadir in the second trimester.

#423 · Systemic Hypertension and Hypotension

Which two adverse effects distinguish ACE inhibitors from ARBs?

Answer

Dry cough and angioedema, both bradykinin-mediated, are far more common with ACE inhibitors. Hyperkalemia and creatinine rise occur with both classes.

#424 · Heart Failure and Cardiomyopathy

What is the target daily dose of metoprolol succinate in HFrEF?

Answer

200 mg once daily, uptitrated from 12.5 to 25 mg daily as tolerated.

#425 · Arrhythmias

How does the PR interval behave in Mobitz type I versus Mobitz type II second-degree AV block?

Answer

Mobitz I: progressive PR prolongation before a dropped beat (usually AV nodal, benign). Mobitz II: fixed PR with sudden nonconducted P waves (infranodal, pacemaker indicated).

#426 · Heart Failure and Cardiomyopathy

What is the target daily dose of eplerenone in heart failure therapy?

Answer

50 mg once daily, started at 25 mg daily. Eplerenone causes less gynecomastia than spironolactone.

#427 · Pulmonary Circulation Disorders

When is systemic thrombolysis clearly indicated in acute pulmonary embolism?

Answer

High-risk PE with hemodynamic instability and acceptable bleeding risk. Intermediate-risk patients get anticoagulation with monitoring, reserving rescue thrombolysis for hemodynamic deterioration.

#428 · Arrhythmias

In wide-complex tachycardia, what single ECG finding proves ventricular tachycardia when present?

Answer

AV dissociation, including capture beats and fusion beats: independent atrial activity confirms a ventricular origin. Its absence does not exclude VT.

#429 · Valvular Disease

List the five minor Duke criteria for infective endocarditis.

Answer

Predisposing condition or injection drug use, fever ≥ 38.0 C, vascular phenomena, immunologic phenomena, and microbiologic evidence not meeting a major criterion.

#430 · Heart Failure and Cardiomyopathy

Roughly how much does an intra-aortic balloon pump augment cardiac output, and what are its main effects?

Answer

About 0.5 L/min. It inflates in diastole to augment coronary perfusion and deflates in systole to reduce afterload.

#431 · Valvular Disease

Should aspirin be added routinely to warfarin for every mechanical valve patient?

Answer

No. The 2020 guideline dropped routine aspirin addition because of bleeding risk; add it only when another indication such as coronary disease exists.

#432 · Congenital Heart Disease

Who qualifies for PFO closure after stroke, based on trials like CLOSE and RESPECT?

Answer

Patients roughly 18 to 60 years old with an embolic-appearing cryptogenic stroke and no other identified cause, after shared decision making with neurology; closure beats antiplatelets alone.

#433 · Arrhythmias

What accessory pathway property on invasive testing marks high sudden death risk in WPW?

Answer

A shortest preexcited RR interval ≤ 250 ms during atrial fibrillation, indicating rapid antegrade conduction; ablation is recommended. Abrupt delta wave loss with exercise suggests lower risk.

#434 · Arrhythmias

AF of unknown or ≥ 48 hours duration: what are the two acceptable pathways to elective cardioversion?

Answer

Either 3 weeks of continuous therapeutic anticoagulation first, or a TEE excluding left atrial appendage thrombus, then cardiovert on anticoagulation.

#435 • Coronary Artery Disease

ST elevation in leads II, III, and aVF localizes to which wall and which artery?

Answer

Inferior wall. The right coronary artery is the culprit in about 80 to 90%, the left circumflex in the remainder.

#436 • Pericardial Disease

Which biologic agents treat colchicine-resistant, steroid-dependent recurrent pericarditis?

Answer

Interleukin-1 blockers: rilonacept, shown to prevent recurrences in the RHAPSODY trial, and anakinra.

#437 · Arrhythmias

What ECG criteria define multifocal atrial tachycardia, and what is the treatment focus?

Answer

Irregular atrial tachycardia with ≥ 3 distinct P wave morphologies and varying PR intervals, classically in decompensated COPD. Treat the underlying illness; cardioversion is ineffective.

#438 · Arrhythmias

Above what PVC burden on 24-hour monitoring does PVC-induced cardiomyopathy become a real concern?

Answer

Roughly ≥ 15 to 20% of total beats. Falling EF with a high burden warrants suppression with beta blockers, antiarrhythmics, or catheter ablation, which can normalize EF.

#439 · Pulmonary Circulation Disorders

What test is mandatory to confirm pulmonary arterial hypertension before starting PAH-specific therapy?

Answer

Right heart catheterization. It confirms precapillary hemodynamics, excludes left heart disease, and allows vasoreactivity testing in eligible patients.

#440 · Arrhythmias

Who is a candidate for percutaneous left atrial appendage occlusion in atrial fibrillation?

Answer

Patients with elevated stroke risk who have a legitimate long-term contraindication to oral anticoagulation, such as recurrent major bleeding; devices include Watchman and Amulet.

#441 · Heart Failure and Cardiomyopathy

How is iron deficiency defined in heart failure for treatment purposes?

Answer

Ferritin <100 ng/mL, or ferritin 100 to 299 ng/mL with transferrin saturation <20%.

#442 · Congenital Heart Disease

What late problems follow atrial switch (Mustard or Senning) repair of D-transposition?

Answer

Failure of the systemic right ventricle with tricuspid regurgitation, baffle leaks or stenosis, sinus node dysfunction, and atrial arrhythmias.

#443 · Arrhythmias

Which two AF populations must receive warfarin rather than a DOAC for stroke prevention?

Answer

Patients with a mechanical heart valve and those with moderate to severe (rheumatic) mitral stenosis. DOACs are contraindicated in both; dabigatran caused harm in RE-ALIGN.

#444 · Coronary Artery Disease

HEART score strata: what ranges define low, moderate, and high risk?

Answer

0 to 3 is low risk with about 1.7% 6-week MACE, suitable for early discharge; 4 to 6 is moderate; 7 to 10 is high risk.

#445 • Systemic Hypertension and Hypotension

What office blood pressure defines normal in the ACC/AHA classification?

Answer

Below 120/80 mmHg. Both the systolic and diastolic values must be below threshold to qualify as normal.

#446 • Coronary Artery Disease

Ticagrelor dosing in ACS and its two key contraindications or cautions?

Answer

180 mg load, then 90 mg twice daily.
Contraindicated with prior intracranial hemorrhage and severe hepatic impairment, and maintenance aspirin must stay ≤ 100 mg daily.

#447 · Arrhythmias

What resting heart rate target defines lenient rate control in AF, and which trial supports it?

Answer

Resting rate under 110 bpm in patients with preserved EF and acceptable symptoms, based on RACE II, which found lenient control noninferior to strict control.

#448 · Arrhythmias

Why does the 2023 AF guideline favor catheter ablation in AF patients with HFrEF?

Answer

Trials such as CASTLE-AF showed ablation reduces mortality and heart failure hospitalization versus medical therapy in selected HFrEF patients (Class 1 recommendation).

#449 · Pericardial Disease

What mitral inflow Doppler finding supports tamponade physiology on echocardiography?

Answer

Exaggerated respiratory variation of mitral inflow velocity, greater than about 30%, with an even larger tricuspid inflow variation above about 60%, reflecting ventricular interdependence.

#450 · Vascular Diseases

Who gets one-time ultrasound screening for abdominal aortic aneurysm?

Answer

Men aged 65 to 75 who have ever smoked, per USPSTF. Selective screening applies to never-smoking men in that age range.

#451 · Vascular Diseases

What is the elective repair threshold for a sporadic ascending thoracic aortic aneurysm?

Answer

≥5.5 cm in the absence of a genetic aortopathy, with lower thresholds for rapid growth or planned concomitant cardiac surgery.

#452 · Arrhythmias

When is catheter ablation a first-line Class 1 rhythm-control option in atrial fibrillation, per 2023 guidance?

Answer

In symptomatic paroxysmal AF in younger patients with few comorbidities, to improve symptoms and reduce progression, without requiring a failed antiarrhythmic drug first.

#453 · Systemic Hypertension and Hypotension

What systolic and diastolic readings define stage 2 hypertension?

Answer

Systolic ≥ 140 mmHg or diastolic ≥ 90 mmHg.

#454 · Arrhythmias

Beyond magnesium, name two therapies for recurrent torsades from acquired QT prolongation with bradycardia.

Answer

Temporary transvenous overdrive pacing (around 100 bpm) or isoproterenol to raise heart rate and shorten the QT, plus potassium repletion toward high normal and stopping QT-prolonging drugs.

#455 · Arrhythmias

How does starting amiodarone affect warfarin and digoxin dosing?

Answer

Amiodarone inhibits their metabolism and transport: reduce warfarin dose by roughly one third to one half and halve digoxin, then follow INR and digoxin levels.

#456 · Heart Failure and Cardiomyopathy

What separates NYHA class II from NYHA class III heart failure symptoms?

Answer

Class II means symptoms with ordinary activity; class III means symptoms with less than ordinary activity, with comfort only at rest.

#457 · Arrhythmias

Describe the pill-in-pocket strategy for infrequent symptomatic AF episodes.

Answer

A single self-administered oral dose of flecainide or propafenone at AF onset, in patients without structural heart disease, after first proving safety in a monitored setting.

#458 · Heart Failure and Cardiomyopathy

What BNP level effectively rules out acute heart failure in a dyspneic patient?

Answer

BNP <100 pg/mL makes acute heart failure unlikely in the acutely dyspneic patient.

#459 · Normal Cardiovascular Anatomy and Physiology

Which ECG leads reflect the inferior wall, and which artery usually supplies it?

Answer

Leads II, III, and aVF. The right coronary artery supplies the inferior wall in the roughly 85% of people with right-dominant circulation.

#460 · Systemic Hypertension and Hypotension

What reading pattern defines masked hypertension, and why does it matter clinically?

Answer

Normal office readings with elevated out-of-office readings. It carries cardiovascular risk similar to sustained hypertension, so it warrants treatment.

#461 • Coronary Artery Disease

STEMI criteria in leads V2 and V3: what ST-elevation thresholds apply by age and sex?

Answer

≥2 mm in men ≥40 years, ≥2.5 mm in men under 40, and ≥1.5 mm in women, measured at the J point.

#462 • Coronary Artery Disease

What is the current role of glycoprotein IIb/IIIa inhibitors in PCI, and their signature adverse effects?

Answer

Bailout use for large thrombus, no-reflow, or thrombotic complications, not routine upstream therapy. Key risks are bleeding and acute profound thrombocytopenia.

#463 · Arrhythmias

What everyday state can unmask a Brugada pattern and must be treated aggressively?

Answer

Fever unmasks the type 1 pattern and precipitates arrhythmias; treat with antipyretics promptly. Sodium channel blockers, cocaine, and heavy alcohol should also be avoided.

#464 · Valvular Disease

What are the two major Duke criteria for infective endocarditis?

Answer

Positive blood cultures with typical organisms or persistent bacteremia, and imaging evidence of endocardial involvement such as vegetation, abscess, prosthetic dehiscence, or new valvular regurgitation.

#465 · Heart Failure and Cardiomyopathy

State normal right atrial pressure, pulmonary capillary wedge pressure, and cardiac index values.

Answer

Right atrial pressure 2 to 6 mmHg, wedge pressure 8 to 12 mmHg, and cardiac index 2.5 to 4.0 L/min/m².

#466 · Arrhythmias

A patient has unexplained syncope and bifascicular block on ECG. What diagnosis must you presume?

Answer

Intermittent complete heart block. Evaluation (or empiric pacing when EP study shows a prolonged HV interval, or per guideline in select cases) is required; do not dismiss as vasovagal.

#467 · Heart Failure and Cardiomyopathy

Which DPP-4 inhibitors are associated with increased heart failure hospitalization?

Answer

Saxagliptin and alogliptin. Saxagliptin increased heart failure hospitalization in SAVOR-TIMI 53, so both carry heart failure warnings.

#468 · Valvular Disease

How is prosthesis type chosen for patients aged 50 to 65?

Answer

Either mechanical or bioprosthetic is reasonable; the choice is individualized through shared decision-making about anticoagulation, reoperation risk, and preference.

#469 · Arrhythmias

Which vagal maneuver technique has the best evidence for terminating SVT, per the REVERT trial?

Answer

The modified Valsalva: 15 second strain, then immediately lie supine with legs raised. It roughly quadrupled conversion versus standard Valsalva.

#470 · Systemic Hypertension and Hypotension

What is the preferred fourth agent for resistant hypertension?

Answer

Spironolactone, shown in PATHWAY-2 to outperform doxazosin and bisoprolol as add-on therapy. Monitor potassium.

#471 • Coronary Artery Disease

Fourth Universal Definition: what two elements are required to diagnose acute myocardial infarction?

Answer

A rise and/or fall in cardiac troponin with at least one value above the 99th percentile upper reference limit, plus clinical evidence of acute myocardial ischemia.

#472 • Systemic Hypertension and Hypotension

Name four elements of correct office blood pressure measurement technique.

Answer

Seated quietly 5 minutes, back supported and feet flat; arm supported at heart level; correct cuff size; average of at least two readings, no caffeine or smoking within 30 minutes.

#473 · Systemic Hypertension and Hypotension

Which initial classes work best in Black adults without heart failure or CKD?

Answer

A thiazide diuretic or a calcium channel blocker, which lower blood pressure more effectively than ACE inhibitor monotherapy in this group.

#474 · Valvular Disease

When is warfarin fetotoxic in pregnancy, and how is anticoagulation handled around delivery?

Answer

Embryopathy risk is greatest during weeks 6 to 12; before planned vaginal delivery, convert to intravenous unfractionated heparin, stopped 4 to 6 hours before delivery.

#475 · Systemic Disorders Affecting the Circulatory System

Which common chemotherapy agents cause coronary vasospasm, and how does it present?

Answer

Fluorouracil (5-FU) and its oral prodrug capecitabine. They can cause chest pain with ischemic ECG changes during infusion, treated by stopping the drug and giving calcium channel blockers or nitrates.

#476 · Heart Failure and Cardiomyopathy

What LVOT gradient thresholds define obstruction in HCM at rest and with provocation?

Answer

A peak gradient ≥ 30 mmHg defines obstruction; resting or provoked gradients ≥ 50 mmHg mark the threshold for septal reduction therapy in refractory symptomatic patients.

#477 · Valvular Disease

How should the TAVR versus SAVR decision be made for patients aged 65 to 80?

Answer

Either option is acceptable; the guideline calls for shared decision-making weighing anatomy, valve durability, access, and patient preference.

#478 · Coronary Artery Disease

Which stress modality is preferred in left bundle branch block, and why?

Answer

Vasodilator nuclear perfusion imaging. Exercise or dobutamine causes false-positive septal perfusion defects from rate-related abnormal septal motion in LBBB.

#479 · Arrhythmias

Describe the ECG signature and drug sensitivity of idiopathic right ventricular outflow tract VT.

Answer

LBBB morphology with an inferior (tall inferior lead) axis in a structurally normal heart; it is adenosine and verapamil sensitive and often triggered by exercise. Ablation is curative.

#480 · Arrhythmias

Name two classic long RP tachycardias to consider when the P wave sits closer to the next QRS.

Answer

Atrial tachycardia and atypical AVNRT; also PJRT from a slowly conducting accessory pathway. Sinus tachycardia is also long RP.

#481 • Coronary Artery Disease

Aspirin in suspected ACS: what loading dose and how should it be taken?

Answer

162 to 325 mg of non-enteric-coated aspirin, chewed for rapid absorption, then 81 mg daily indefinitely.

#482 • Coronary Artery Disease

Clopidogrel with fibrinolysis for STEMI: how does loading differ by age?

Answer

Patients ≤ 75 years get a 300 mg load; patients over 75 get no load, only 75 mg daily, to limit intracranial bleeding risk.

#483 · Heart Failure and Cardiomyopathy

State the oral loop diuretic dose equivalence for furosemide, bumetanide, and torsemide.

Answer

Furosemide 40 mg equals bumetanide 1 mg equals torsemide 20 mg by mouth.

#484 · Systemic Hypertension and Hypotension

What blood pressure change on standing defines orthostatic hypotension?

Answer

A sustained fall of ≥ 20 mmHg systolic or ≥ 10 mmHg diastolic within 3 minutes of standing or head-up tilt.

#485 · Normal Cardiovascular Anatomy and Physiology

How are preload and afterload defined in ventricular mechanics?

Answer

Preload is ventricular wall stress at end-diastole, approximated by end-diastolic volume or filling pressure. Afterload is the wall stress the ventricle must overcome during ejection.

#486 · Congenital Heart Disease

After an arterial switch operation for D-TGA, what anatomy needs lifelong surveillance?

Answer

The reimplanted coronary arteries (ostial stenosis causing ischemia), the neo-aortic root (dilation and regurgitation), and the branch pulmonary arteries (supravalvular stenosis).

#487 · Pericardial Disease

Which echocardiographic and Doppler signs point to constrictive pericarditis?

Answer

Respirophasic septal shift with a septal bounce, annulus reversus (lateral e' lower than medial e'), and expiratory diastolic flow reversal in the hepatic veins.

#488 · Coronary Artery Disease

Why is P2Y12 inhibitor monotherapy, rather than aspirin monotherapy, the usual agent kept after abbreviated DAPT?

Answer

Trials of early aspirin withdrawal, such as TWILIGHT, continued the P2Y12 inhibitor and showed less bleeding with no increase in ischemic events.

#489 · Arrhythmias

State the strongest (Class 1) indication for cardiac resynchronization therapy.

Answer

LVEF $\leq 35\%$, sinus rhythm, LBBB with QRS ≥ 150 ms, and NYHA class II to IV symptoms on GDMT. Benefit falls with non-LBBB morphology or narrower QRS.

#490 · Vascular Diseases

What is cilostazol's role in claudication and its absolute contraindication?

Answer

It improves pain-free walking distance. It is contraindicated in heart failure of any severity because oral phosphodiesterase-3 inhibitors increase mortality in that setting.

#491 • Coronary Artery Disease

Inclisiran: what is its mechanism and its maintenance dosing schedule?

Answer

A small interfering RNA that silences hepatic PCSK9 synthesis. Dosed at baseline, 3 months, then every 6 months, lowering LDL-C by about 50%.

#492 • Coronary Artery Disease

What coronary artery calcium findings favor initiating statin therapy in borderline or intermediate risk patients?

Answer

A CAC score ≥ 100 Agatston units or ≥ 75 th percentile for age and sex. Any score of 1 to 99 favors a statin, particularly after age 55.

#493 · Heart Failure and Cardiomyopathy

What is the evidence-based route of iron repletion in iron-deficient HFrEF?

Answer

Intravenous iron, such as ferric carboxymaltose, improves symptoms, functional capacity, and quality of life (FAIR-HF, AFFIRM-AHF). Oral iron failed to replete iron or improve capacity (IRONOUT HF).

#494 · Arrhythmias

State the primary prevention ICD criteria for ischemic cardiomyopathy with NYHA class II or III symptoms.

Answer

LVEF $\leq 35\%$ on GDMT, at least 40 days after MI and 90 days after revascularization, with >1 year meaningful survival expected.

#495 · Arrhythmias

Who with Brugada syndrome gets an ICD, and what drug helps recurrent arrhythmic storms?

Answer

ICD for survivors of cardiac arrest or arrhythmic syncope with a spontaneous type 1 pattern. Quinidine (or catheter ablation of the RVOT epicardium) treats recurrent VF or storm.

#496 · Arrhythmias

What safety rules govern sotalol initiation and ongoing use?

Answer

It is renally cleared and prolongs the QT: check creatinine, avoid significant renal failure, and monitor QTc during initiation (classically in-hospital for 3 days), especially with hypokalemia or bradycardia.

#497 · Coronary Artery Disease

Name the seven variables of the TIMI risk score for NSTEMI-ACS.

Answer

Age ≥ 65 , ≥ 3 CAD risk factors, known stenosis $\geq 50\%$, aspirin use in the prior 7 days, ≥ 2 anginal episodes in 24 hours, ST deviation ≥ 0.5 mm, and elevated biomarkers.

#498 · Heart Failure and Cardiomyopathy

Which drug class first showed convincing outcome benefit in HFrEF, and on what evidence?

Answer

SGLT2 inhibitors, based on EMPEROR-Preserved and DELIVER, which reduced heart failure hospitalization and cardiovascular death; class 2a in the 2022 guideline.

#499 · Coronary Artery Disease

What is the door-to-needle goal for fibrinolytic therapy in STEMI?

Answer

30 minutes or less from hospital arrival to fibrinolytic administration.

#500 · Valvular Disease

Which criterion combinations classify endocarditis as definite by Duke criteria?

Answer

Two major criteria, one major plus three minor, or five minor criteria.